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Carboniferous palaeomagnetism of the Rocky Creek Block, northern Tamworth Belt, and the New England pole path*

C. T. KLOOTWIJK†

Australian Geodynamics Cooperative Research Centre, Geoscience Australia, GPO Box 378, Canberra, ACT 2601, Australia.

Palaeomagnetic, rock magnetic and magnetic fabric results are presented for a Carboniferous (Visean to Westphalian) succession of felsic, mainly ignimbritic, volcanic and volcaniclastic rocks from the Rocky Creek Block of the northern Tamworth Belt, southern New England Orogen. Detailed thermal demagnetisation of 734 samples from 64 sites show three groups of magnetic components with low (<300°C), intermediate (300–600°C) and high (500–680°C) unblocking temperature ranges. Well-defined primary magnetisations have been determined for 28 sites with evidence of four overprint phases. The overprints arise from a mid-Tertiary weathering event (or possibly recent viscous origin), and from fluid movements associated with the Late Cretaceous opening of the Tasman Sea, thrusting during the Middle Triassic main phase of the Hunter–Bowen Orogeny, and latest Carboniferous – Early Permian formation of the Bowen–Gunnedah–Sydney Basin system. Rock magnetic tests establish that the primary magnetisation carriers in the volcanic rocks are mainly magnetite (predominantly single domain, or pseudo-single domain, and little or no multidomain) and hematite. Optimal magnetic cleaning is achieved at high to very high temperatures, with subtle, but systematic, directional and statistical differences between primary components derived from the mainly hematite fraction and pseudo-components derived from the mainly magnetite fraction. The 28 primary magnetisation results are presented as six mean-site results, summarised below and representing 25 sites, and three single-site results. Fold tests could be applied to five mean-site results. These are all positive, but one of these results may represent a secondary magnetisation. The primary magnetisation results define a Visean to Westphalian pole path. This long pole path indicates extensive latitudinal and rotational movement for the Rocky Creek Block, and potentially for the New England Orogen, as follows: (i) Yuendoo Rhyolite Member (Caroda Formation, Visean) pole 235.8°E, 27.7°S, ED_{95} = 9.0°, n = 3; (ii) Peri Rhyolite Member/Boomi Rhyolite Member (Clifden Formation, Namurian, 318.0 ± 3.4 Ma) pole 177.4°E, 63.4°S, ED_{95} = 5.2°, n = 3; (iii) tuffaceous beds above Boomi Rhyolite Member (Clifden Formation?, Namurian) pole 162.2°E, 59.1°S, ED_{95} = 10.2°, n = 3; ((iv) upper Clifden Formation/lower Rocky Creek Conglomerate (Namurian/Westphalian) pole 95.3°E, 49.6°S, ED_{95} = 8.1°, n = 3 (possible overprint)); (v) Rocky Creek Conglomerate (Westphalian) pole 136.5°E, 57.6°S, ED_{95} = 5.3°, n = 5; (vi) Lark Hill Formation (Westphalian) pole 127.0°E, 50.4°S, ED_{95} = 4.8°, n = 8.

KEY WORDS: Carboniferous, New England Orogen, palaeomagnetism, pole path, Rocky Creek Block, Tamworth Belt.

INTRODUCTION

This is the first of a number of detailed palaeomagnetic studies on Carboniferous – Early Permian successions of the Tamworth Belt in the southern New England Orogen, covering the Rocky Creek, Werrie, Rouchel and Gresford Blocks (Figure 1). The main objectives are to: (i) test on oroclinal bending of the Tamworth Belt; (ii) define a Late Palaeozoic apparent polar wander path for the New England Orogen; (iii) compare this pole path with cratonic Australian pole paths; and (iv) contribute to definition of a representative pole path for Australia and (eastern) Gondwanaland.

An extensive set of palaeomagnetic data for the Late Palaeozoic of Australia has been accumulated over the past four decades and is now available through the global palaeomagnetic database (McElhinny & Lock 1996), with additional preliminary results in list form (Klootwijk 1996a). However, there is still no consensus on the shape of Australia's Late Palaeozoic pole path. Definition has suffered among others from general absence of a middle Carboniferous stratigraphic record in cratonic sedimentary basins and in the Tasmanides, from severe Early

*Appendix 1 and Tables 5–7 [indicated by an asterisk (*) in the text and listed at the end of the paper] are Supplementary Papers lodged with the National Library of Australia (Manuscript Section); copies may be obtained from the Geological Society of Australia website at www.gsa.org.au

†Present address: Department of Geology, Australian National University, Canberra, ACT 0200, Australia (Chris.Klootwijk@geology.anu.edu.au).

and middle Carboniferous overprinting related to the Kanimblan and Alice Springs Orogenies, and from latest Carboniferous – Early Permian overprinting probably related to formation of the Bowen–Gunnedah–Sydney Basin system.

The Carboniferous – Early Permian succession of the Tamworth Belt has several features favourable to defining Australia's Late Palaeozoic pole path, with others adding complexity. The frequent presence of felsic volcanic rocks, predominantly ignimbrites, favours determination of primary magnetisations and SHRIMP U–Pb zircon ages. Complexities such as, time of accretion, displacement with respect to the Australian craton associated with formation of the Bowen–Gunnedah–Sydney Basin system and rotational deformation associated with the Texas – Coffs Harbour and Manning Oroclines need resolving before Tamworth Belt palaeomagnetic data can be applied to define the Australian pole path.

GEOLOGY

New England Orogen: overview

The New England Orogen is the easternmost, youngest accreted segment of the Tasman Orogenic System or Tasmanides, and forms an elongate belt stretching for more than 2000 km from Townsville to Newcastle with a width of up to 500 km. It is traditionally divided into a northern and southern part separated by the Mesozoic Surat and Clarence–Moreton Basins: i.e. the northern New England Orogen or Yarrol Orogen and the southern New England Orogen or New England Orogen proper. The southern New England Orogen is characterised by oroclines (Harrington & Korsch 1985a): the Texas and Coffs Harbour Oroclines in the north and the Manning Orocline in the south (Figure 1b). An oroclinal origin for the latter has been questioned (Lennox *et al.* 1997). The time of their formation is also under dispute, but an Early Permian origin is most likely (Harrington & Korsch 1985a; Korsch & Harrington 1987). The Bowen–Gunnedah–Sydney Basin system is generally interpreted as a foreland basin to the New England Orogen. The Tamworth Belt succession overthrusted the Bowen–Gunnedah–Sydney Basin system westward during the Middle Triassic main phase of the Hunter–Bowen Orogeny (Harrington & Korsch 1985a, b; Harrington 1989; Veevers *et al.* 1994a; Roberts & Geeve 1999), probably related to accretion of the Gympie Terrane, the easternmost part of the New England Orogen (Harrington & Korsch 1985a; Harrington 1989).

The stratigraphy of the New England Orogen has been traced back to the Early Cambrian, although preservation of its early stratigraphic record is limited. The stratigraphic succession is largely made up of Middle and Upper Palaeozoic marine sediments with widespread magmatic activity characterised by Carboniferous felsic volcanics and Permo-Triassic granites and felsic volcanics. Various tectonostratigraphic subdivisions of the New England Orogen have been proposed (Leitch 1974; Korsch 1977; Roberts & Engel 1980; Cawood & Leitch 1985). The gross, fourfold, subdivisions by Coney *et al.* (1990) and Scheibner and Basden (1998) and the more detailed subdivision by Aitchison and Flood (1995) are followed here (Figure 2). The New England Orogen, or superterrane, can be divided from west to east into the Proserpine (Connors–Auburn) magmatic belt, the Yarrol–Tamworth Terrane(s), the Wandilla–Gwydir Terrane (Wandilla Tablelands Terrane: Scheibner & Basden 1998), and the Nambucca and Gympie Terranes. Palaeogeographically, and from west to east, these tectonostratigraphic terranes have been identified as a magmatic

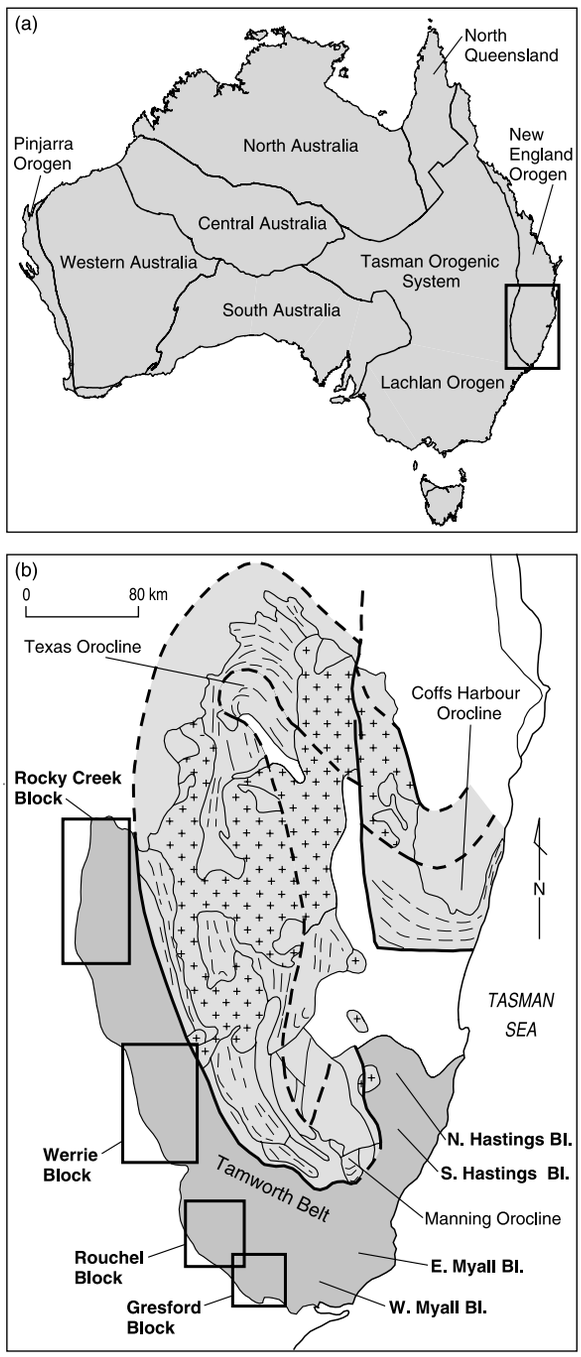


Figure 1 (a) Australian mega-elements, representing continent-scale groups of crustal elements (after Shaw *et al.* 1995 figure 3). Rectangle outlines area in (b). (b) Schematic overview of the southern New England Orogen (after Korsch & Harrington 1987 figure 2) with outlines of the four studied tectonic units in the Tamworth Belt (dark-grey shading).

arc (Connors–Auburn), a volcanic arc–forearc (Yarrol–Tamworth), an accretionary wedge (Wandilla–Gwydir), a marginal sea (Nambucca), and an exotic volcanic arc (Gympie).

Tamworth Belt: Upper Palaeozoic succession

The Tamworth Belt (Korsch 1977), the southern part of the Yarrol–Tamworth Terrane, is characterised by two arc-related successions with distinct tectonostratigraphic development. The older (Silurian?)–Devonian intra-oceanic arc, the Gamilaroi Terrane (Flood & Aitchison 1992; Aitchison & Flood 1995 figure 1), may already have been

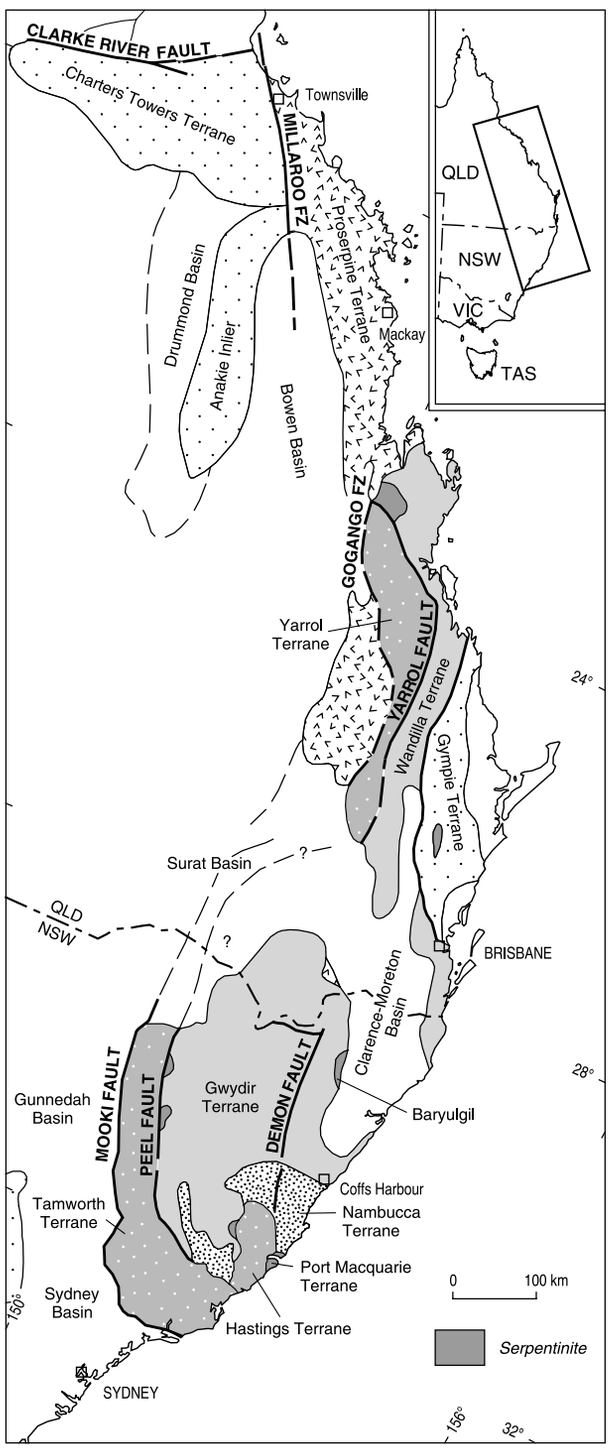


Figure 2 Terranes in the New England Orogen (after Coney *et al.* 1990 figure 11).

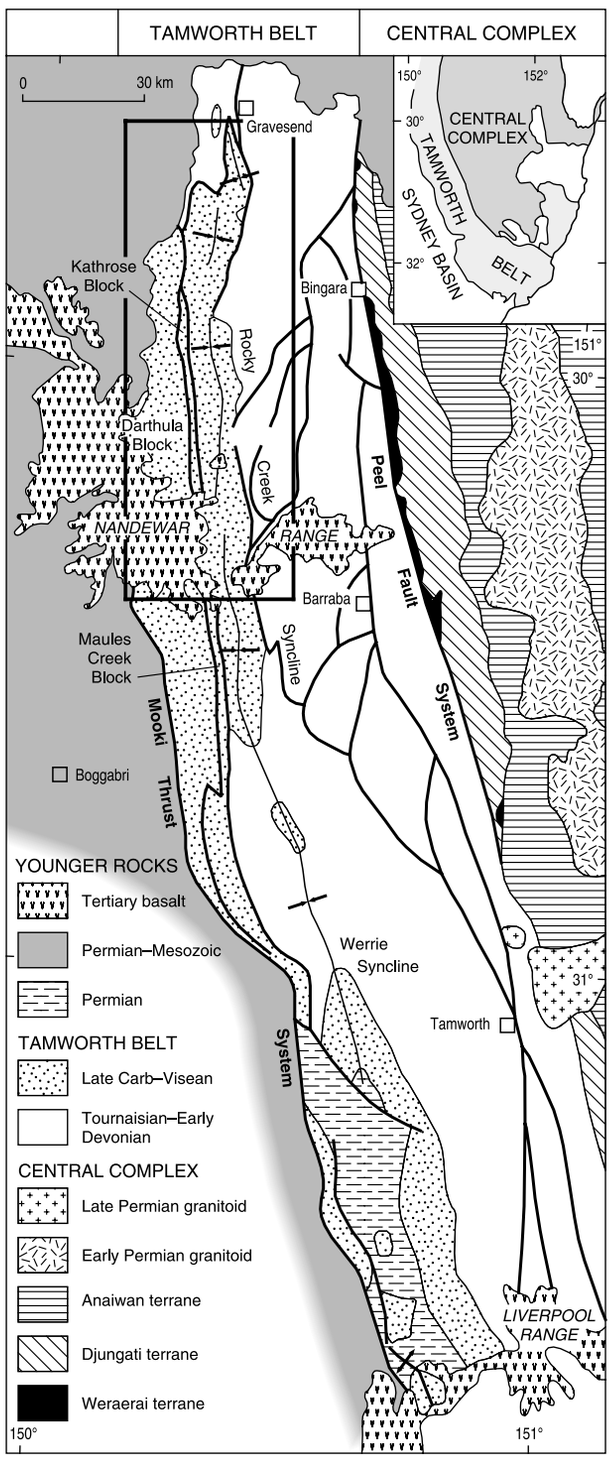


Figure 3 Regional geological map of the northern Tamworth Belt (after Opdyke *et al.* 2000 figure 1; Pogson 1972). Rectangle shows the sampled regions of the Rocky Creek, Kathrose and Darthula Thrust Sheets in the far northern part of the Tamworth Belt. Terranes of the Central Complex are indicated (after Flood & Aitchison 1988 figure 1)

accreted to the cratonic Lachlan Orogen by the Late Devonian (Famennian) and most probably by the middle Carboniferous (Powell *et al.* 1990). The younger succession is mainly made up of Upper Devonian to Carboniferous volcanic and volcaniclastic rocks with shelf-like characteristics, deepening toward the east. The western continental-arc source, the hypothetical Baldwin Arc (Veevers *et al.* 1982; McPhie 1987; Coney *et al.* 1990; Scheibner & Basden 1998; Roberts & Geeve 1999; Opdyke *et al.* 2000), evolved from andesitic during the Devonian, into dacitic and then rhyolitic compositions during the Carboniferous. The location of the arc has not been established in outcrop. It may have been westward overthrust by the Tamworth Belt succession during the Middle Triassic main phase of the Hunter–Bowen Orogeny, or may have been displaced by dextral strike-slip in the latest Carboniferous – Early Permian (Harrington & Korsch 1985a; Harrington 1989; Korsch *et al.* 1997; Klootwijk *et al.* in press a).

The younger forearc basin succession starts with uppermost Devonian deep-marine sediments, overlain by a Lower Carboniferous shelfal succession with volcanic intercalations in its upper part. The upper Visean – lower Namurian is characterised by a disconformity spanning a hiatus of variable, up to 20 million years, duration along the belt (Opdyke *et al.* 2000). From middle to late Visean onward the succession contains frequent felsic volcanic (e.g. ignimbrites, ash-fall tuffs, lavas) and volcaniclastic rocks (e.g. volcanogenically derived sandstones to mudstones). Local biostratigraphic control on the Carboniferous succession is limited and globally correlatable faunal control is sparse or unavailable. The widely occurring felsic volcanics are prospective for SHRIMP U–Pb zircon studies and have been extensively dated (Claoué-Long *et al.* 1995; Opdyke *et al.* 2000; Roberts *et al.* 1995a, 1995b). Compartmentalisation of this younger succession has been recognised (Roberts & Engel 1987; Roberts *et al.* 1991). Roberts and Geeve (1999) identified seven blocks, separated by fault zones/lineaments. From northwest to southeast these are: the Rocky Creek, Werrie, Rouchel, Gresford, Myall, southern Hastings, and northern Hastings Blocks (Figure 1b). Stratigraphic development varies considerably between the blocks and is detailed here for the Rocky Creek Block.

Rocky Creek Block

The Carboniferous succession of the Rocky Creek Block is made up of a lower and an upper part separated by a late Visean to early Namurian hiatus of approximately 10 million years duration according to SHRIMP U–Pb zircon data (Opdyke *et al.* 2000). The lower succession is made up of Tournaisian marine shelf sediments, overlain by Visean volcanogenically derived fluvial sediments with intercalated tuffs and lavas. The upper succession is made up of continental glaciogenic, volcaniclastic and fluvial successions of Namurian and Westphalian to Stephanian age (Opdyke *et al.* 2000).

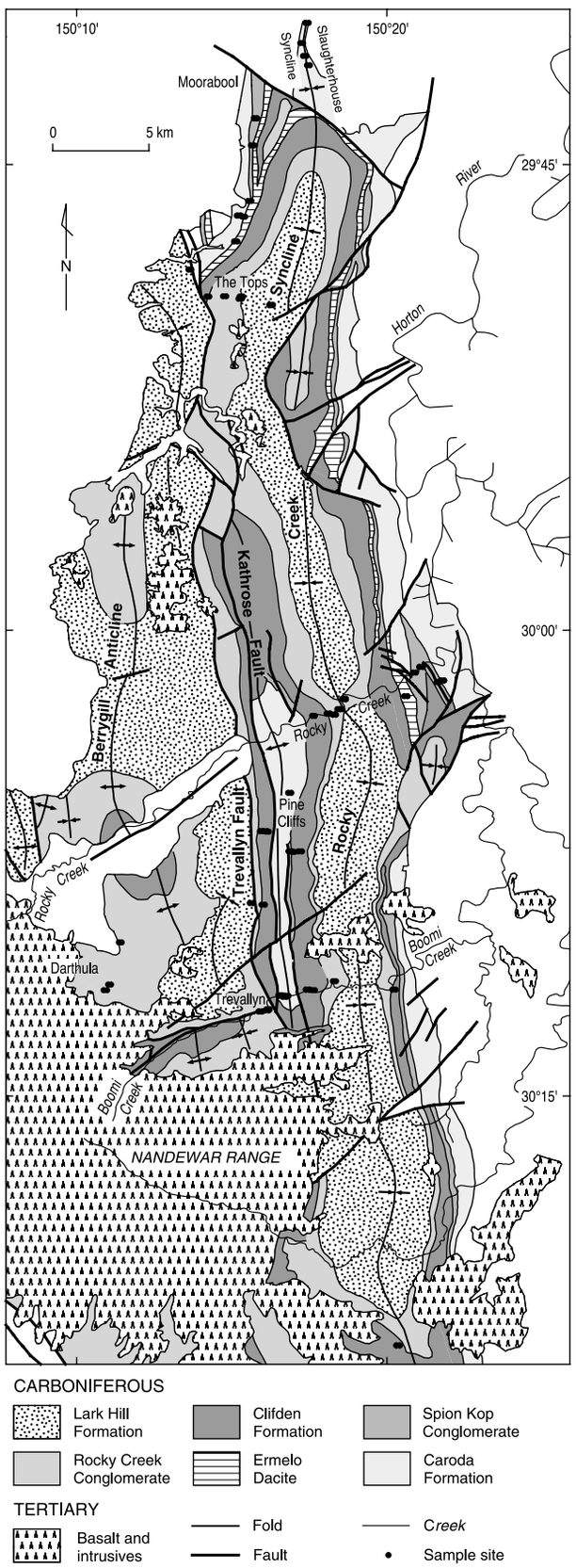


Figure 4 Geological map of Visean to Westphalian formations of the Rocky Creek Block of the northern Tamworth Belt with palaeomagnetic sampling sites. Modified from the Gravesend (Stroud 1990b) and Horton 1:100 000 sheets (Brown 1987), and from Roberts (*in* Opdyke *et al.* 2000 figures 2, 4, 5). Tertiary volcanics and intrusives are shown, but pre-Caroda and post-Lark Hill formations have been omitted for simplicity.

The Carboniferous succession is exposed in three west-vergent thrust sheets, identified here after Opdyke *et al.* (2000), and from west to east, as the Darthula, Kathrose and Rocky Creek Thrust Sheets (Figure 3). The thrust sheets were transported westward (Glen *et al.* 1993; Korsch *et al.* 1997) during the Middle Triassic main phase of the Hunter–Bowen Orogeny over distances up to 75–85 km (Woodward 1995). Shortening estimates vary from at least 10 km (Wartenberg *et al.* 1999) to more than 30 km (Wang & Roberts 1999). Consequently, there is considerable stratigraphic variation between the volcano-sedimentary successions of the three thrust sheets, complicating lithostratigraphic correlations. Recent detailed lithostratigraphic and magnetostratigraphic studies have led to a revised correlation scheme (Opdyke *et al.* 2000) that is followed here, albeit with one proposed refinement. The easternmost thrust sheet has the best exposure and the most extensive outcrop throughout the Rocky Creek Syncline (Figure 4). Its stratigraphy is described here for reference purposes. The Rocky Creek Syncline is separated by the Kathrose Fault from a narrow thrust slice of the Kathrose Thrust Sheet to the west, which itself is separated by the Trevallyn Fault from the westward adjacent, more widely outcropping, Darthula Thrust Sheet (Figure 4).

The lower succession of the Rocky Creek Syncline is made up of the Lower Carboniferous Luton, Namoi and Caroda Formations. The Luton and Namoi Formations are of Tournaisian age in the west, but in the east the Namoi Formation continues through to the middle Visean, representing a local equivalent of the Caroda Formation. Volcanic rocks first appear in the Caroda Formation, with marine faunal evidence for a middle Visean age (Jenkins *et al.* 1993; Roberts *et al.* 1993). The Caroda Formation is the oldest formation sampled (Figure 5). It is characterised in the northern Rocky Creek Block (Yeates 1970; Johnson 1971; Stroud 1990a, b) by two dacitic to rhyodacitic tuffs, the Yuendoo Rhyolite (Opdyke *et al.* 2000) and the West Lynne Rhyodacite (Figure 6). The latter is affected by the erosional disconformity between the lower and upper succession. The Caroda Formation of the central Rocky Creek Syncline is largely devoid of volcanics. A single outcrop only of the lowermost volcanic horizon in the Caroda Formation of the Back Creek region, the High Valley andesite tuff (Hocking 1973), has been located and sampled.

The Spion Kop Conglomerate, the lowest unit of the upper succession, disconformably overlies the Caroda Formation (Figures 5, 6). The glaciogenic conglomerate and diamictite succession of early Namurian age (Opdyke *et al.* 2000) have not been sampled in this study for want of suitable volcanic horizons. The Spion Kop Conglomerate is overlain in the northern part of the Rocky Creek Syncline by the Ermelo Pyroclastics with a hornblende K–Ar age of 319 ± 5 Ma and a SHRIMP U–Pb zircon age of 321.3 ± 3.2 Ma (Figures 5, 6) (Opdyke *et al.* 2000).

The Clifden Formation of middle to late Namurian age (Opdyke *et al.* 2000) overlies the Spion Kop Conglomerate, or the Ermelo Pyroclastics where present (Figures 5, 6).

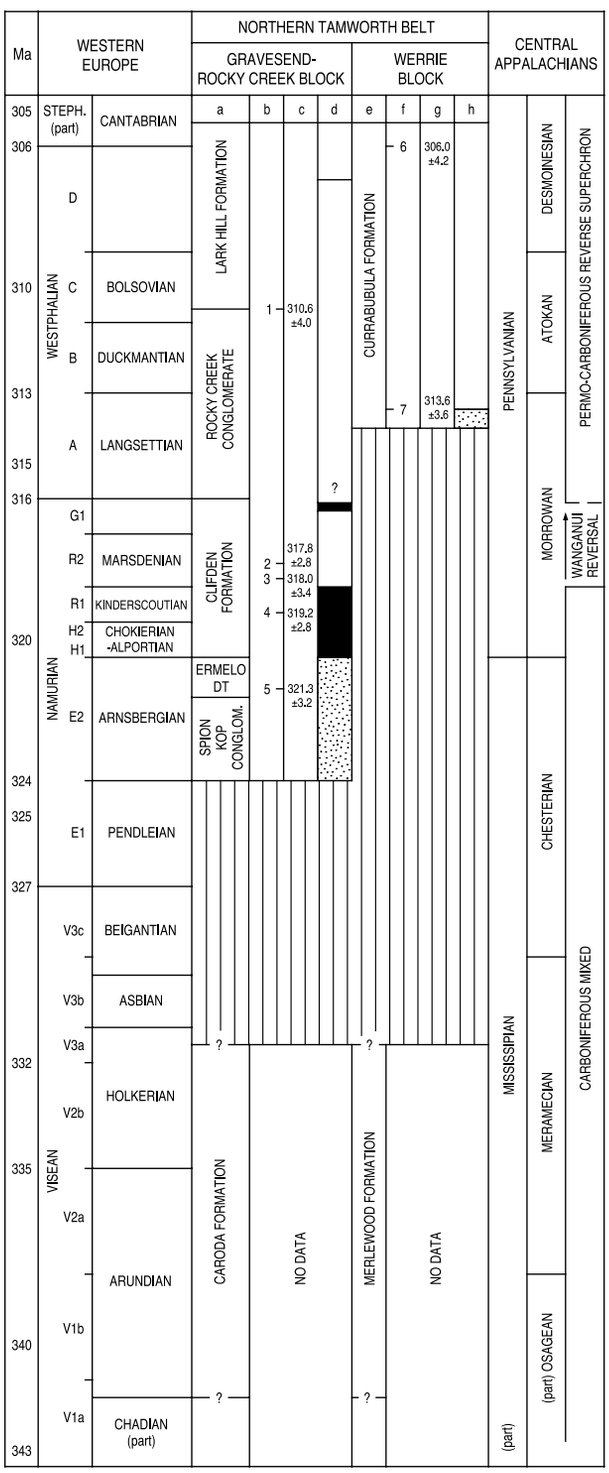


Figure 5 Stratigraphic column for the Rocky Creek and Werrie Blocks of the northern Tamworth Belt in comparison with Carboniferous time units for western Europe and eastern North America. Compiled after Roberts *et al.* (1993, 1995a), Jenkins *et al.* (1993), Jones (1996), Roberts *in* Opdyke *et al.* (2000 figure 18), and P. J. Jones (pers. comm. 1999). a, e, studied stratigraphic succession; b, f, volcanics with SHRIMP U–Pb zircon age control [shown in c and g (Opdyke *et al.* 2000)]; d, h, magnetostratigraphic control (Opdyke *et al.* 2000). Abbreviations in columns b and f: 1, Eulowrie Pyroclastics of the Lark Hill Formation; 2, Eastons Arm Rhyolite; 3, Peri Rhyolite; 4, Wanganui Andesite; 5, Ermelo Pyroclastics; 6, Piallaway Trig Ignimbrite; 7, unnamed ignimbrite at base of Currabubula Formation. In columns d and h: white, reverse polarity; black, normal polarity; stippled, undetermined.

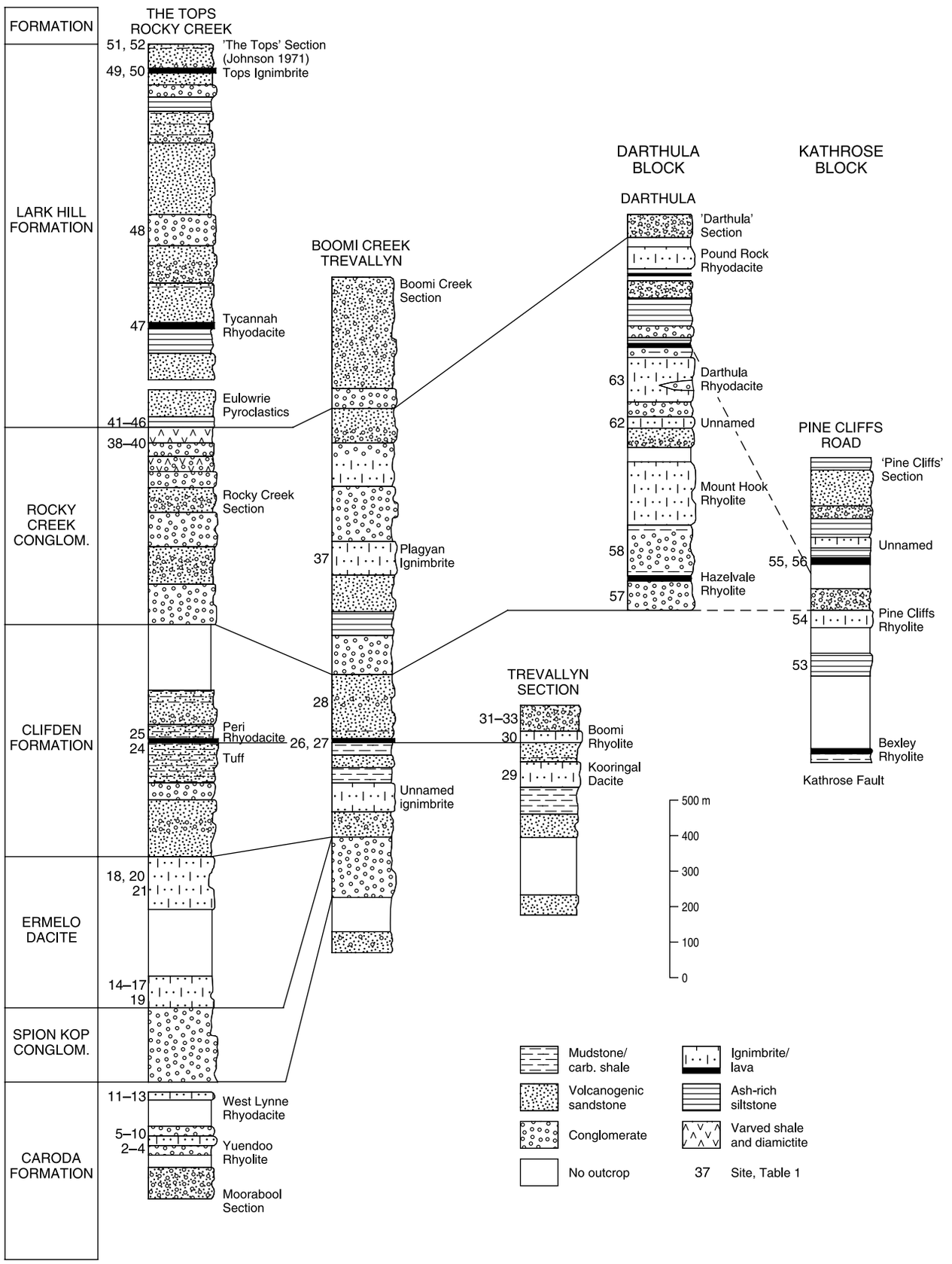


Figure 6 Stratigraphic sections of Visean to Westphalian strata in the Rocky Creek, Kathrose and Darthula Thrust Sheets (modified after Roberts *in* Opdyke *et al.* 2000 figures 4, 5). The stratigraphic position of the Boomi Creek section near Trevallyn has been adapted as discussed in the text. White parts of the column indicate no outcrop. Numbers indicate palaeomagnetic site numbers as in Table 1, but limited to sites actually sampled on, or near to, the profiles shown. The stratigraphic position of sites not shown on the profiles (1, 22, 23, 34–36, 60–61, 64) is described in the text.

It is made up of fluvial feldspatholitic sandstone, plant-bearing siltstone and conglomerate and is characterised in the central Rocky Creek Syncline by the Peri Rhyolite, an ignimbrite with a SHRIMP U–Pb zircon age of 318.0 ± 3.4 Ma (Opdyke *et al.* 2000). The Peri Rhyolite has been correlated with the Eastons Arm Rhyolite of the southern Rocky Creek Syncline (Opdyke *et al.* 2000 figures 4, 5, 11). The Eastons Arm Rhyolite (317.8 ± 2.8 Ma: Opdyke *et al.* 2000) and the underlying Wanganui Andesite (319.2 ± 2.8 Ma: Opdyke *et al.* 2000) are of reverse and normal magnetic polarity, respectively, and have been identified (Opdyke *et al.* 2000) to constrain the lower boundary of the Permo-Carboniferous Reverse Superchron (or 'Kiaman'). This pair of volcanic rocks is of great significance for global correlation and dating. It is discussed herein that the volcanic pair of Boomi Rhyolite and Kooringal Dacite, which crops out in Boomi Creek directly east of the Kathrose Fault (Figures 4, 6), likewise may constrain the lower boundary of the Permo-Carboniferous Reverse Superchron, with the Boomi Rhyolite correlatable with the Peri Rhyolite. In contrast, Opdyke *et al.* (2000), following earlier workers (Hocking 1973; Stroud 1990b), regarded this Boomi Creek succession as a minor thrust sliver belonging to the Caroda Formation. Several regionally less extensive volcanic units occur beneath the Peri Rhyolite in the Back Creek region near Trevallyn (Figure 4): among them the Glen Idle Rhyolite (Hocking 1973; Opdyke *et al.* 2000) and an underlying unnamed vitric volcanic horizon. The Pine Cliffs Road section of the more western Kathrose Thrust Sheet contains only the upper part of the Clifden Formation, and the westernmost Darthula Thrust Sheet contains only its uppermost part (Figure 6).

The Clifden Formation is overlain by the fluvial to distinctly glaciogenic Rocky Creek Conglomerate of early to middle Westphalian age (Figures 5, 6) (Opdyke *et al.* 2000). The glaciogene rocks are largely restricted to the upper part of the formation and crop out well along the Rocky Creek section within the axial region of the central Rocky Creek Syncline. The more prominent volcanic horizon is the Plagyan Ignimbrite (Hocking 1973; Opdyke *et al.* 2000). The Rocky Creek section has a well-exposed succession of vitric tuffs and vitric tuffaceous diamictites in the upper part of the Rocky Creek Conglomerate. The upper part of the Pine Cliffs Road section of the Kathrose Thrust Sheet has been tentatively correlated (Opdyke *et al.* 2000) with the lower part of the Rocky Creek Conglomerate. The westernmost Darthula Thrust Sheet contains various ignimbrites in the Rocky Creek Conglomerate, such as the Hazelvale Rhyolite in the lower part of the formation.

The upper unit of the upper succession, the Lark Hill Formation, has been sampled along Tops Road in the northern Rocky Creek Block and along Rocky Creek in the central Rocky Creek Block (Figure 4). Prominent volcanic horizons on the Tops Road section are the brownish-green Tycannah Rhyodacite (Johnson 1971; Opdyke *et al.* 2000) near its base and the greenish-reddish Tops Ignimbrite (Johnson 1971; Opdyke *et al.* 2000). The Rocky Creek section is characterised by greenish, pinkish ash-rich cherty siltstone of redeposited pyroclastic origin, known as the Eulowrie Pyroclastics (Hocking 1973; Opdyke *et al.* 2000). A SHRIMP U–Pb zircon age of 310.6 ± 4.0 Ma indicates a late Westphalian age (Figures 5, 6). The Eulowrie Pyroclastics and the underlying cherty diamictites of the Rocky Creek Conglomerate were first analysed by Irving (1966) in his pioneering palaeomagnetic study of Carboniferous volcanics of the Tamworth Belt.

Deformation phases

The New England Orogen has a long history of tectonic activity. Up to eight tectonic phases have been identified for the period from Devonian to the Late Cretaceous opening of the Tasman Sea (Korsch & Harrington 1981; Harrington & Korsch 1985a; Veevers *et al.* 1994b; Opdyke *et al.* 2000). The Tamworth Belt has been affected strongly by four of the phases. These may have induced magnetic overprints on the studied successions of the Rocky Creek Block, requiring careful analysis and interpretation of the palaeomagnetic data in order to separate and identify primary and overprint components, and perhaps incompletely cleaned pseudo-components.

The oldest of these four Carboniferous tectonic phases relates to development of the erosional disconformity between the lower and upper succession. Magnetic overprints close to the top of the eroded succession may result from chemical processes during erosion, or may be due to fluid-flow that used the erosional disconformity as a pathway.

The second tectonic phase had a major effect on the shape of the New England Orogen. It involved latest Carboniferous to earliest Permian initiation of the Bowen–Gunnedah–Sydney Basin system by extension and subsequent formation of the Texas – Coffs Harbour and Manning Oroclines. The time of oroclinal deformation is debated (Aubourg *et al.* 1994; Klootwijk in press a), with estimates ranging from latest Carboniferous (Murray *et al.* 1987), earliest Permian (Fergusson & Leitch 1993), to Late Permian (Collins 1991). Some agreement is now emerging for an Early Permian origin as argued by Harrington and Korsch (1985a; Korsch & Harrington 1987; Harrington 1989; Lennox & Flood 1997). Fluid movements related to this major tectonic phase have led to substantial magnetic overprinting with a reverse polarity 'Kiaman' signature (Aubourg *et al.* 1994).

The third phase relates to the Middle Triassic main phase of the Hunter–Bowen Orogeny, when the Tamworth Belt overthrusted westward across the Lachlan Orogen (Korsch *et al.* 1997). Substantial magnetic overprinting related to fluid movements can be expected for this phase.

The fourth phase relates to Late Cretaceous initiation of rifting between Lord Howe Rise and eastern Australia at *ca* 95 Ma and opening of the Tasman Sea at *ca* 84 Ma (Veevers *et al.* 1991; Symonds *et al.* 1996; Müller *et al.* 2000), and is probably also related to a thermal event along the eastern seaboard at *ca* 105–95 Ma, followed by cooling, uplift and denudation at *ca* 95–90 Ma (Schmidt & Embleton 1981; Moore *et al.* 1986; O'Sullivan *et al.* 1999). This opening is characterised by pervasive magnetic overprinting with exclusively normal polarity (Cretaceous Normal Polarity Superchron), widely developed along the eastern seaboard and also recognised inland, along structural zones in particular. Its presence is expected in the Tamworth Belt, being a major zone of prolonged structural weakness.

Table 1 Sample details.

Site no.	Formation/Member/Flow/Site	Acronym	Lat (°S)	Long (°E)	Lithology	Section (cm)	Sample	Initial intensity (mAm^{-1})	Dip direction (°)	Dip (°)
ROCKY CREEK THRUST										
Caroda Formation, central Rocky Creek Block										
1	High Valley andesite tuff[a]	CIBG	30.0872	150.2803	purple ash-fall tuff	45	12	0.6–8.7	103.0	16.1
Caroda Formation, north Rocky Creek Block										
2	~145 m below CIYA	CIYC	29.7775	150.2528	ochre tuffaceous	40	10	4.5–8.6	92.7	57.8
3	~125 m below CIYA	CIYB	29.7773	150.2533	dark grey tuffaceous	150	10	754–2013	92.7	57.8
4	~level of CIYB	CIYH	29.6917	150.2908	dark grey tuffaceous	1 m	12	6.9–1325	262.4	70.8
5	~5 m below AIYB	AIYC	29.7252	150.2619	yellowish tuffaceous	55	10	0.3–13	90.1	56.9
6	Yuendoo Rhyolite[b,c,d]	AIYB	29.7252	150.2619	indurated buff ash-flow tuff	50	10	51–90	90.1	56.9
7	Yuendoo Rhyolite[b,c,d]	CIYA	29.7780	150.2557	indurated ash-flow tuff	170	13	0.3–166	92.7	57.8
8	Yuendoo Rhyolite[b,c,d]	CIYG	29.6967	150.2917	indurated ash-flow tuff	115	11	0.4–1.0	252.1	69.8
9	Yuendoo Rhyolite[b,c,d]	CIYI	29.6848	150.2885	indurated ash-flow tuff	125	11	8.5–30	85.7	54.1
10	Yuendoo Rhyolite[b,c,d]	CIYF	29.6740	150.2913	indurated ash-flow tuff	~50	12	88–855	116.6	50.0
11	West Lynne Rhyodacite[c,d]	AIYA	29.7396	150.2603	red zeolitised ash-fall tuff	185	10	11–31	94.6	61.8
12	West Lynne Rhyodacite[c,d]	CIYE	29.7697	150.2588	red zeolitised ash-fall tuff	140	12	8.1–346	93.4	41.5
13	West Lynne Rhyodacite[c,d]	CIYD	29.7913	150.2513	red zeolitised ash-fall tuff	115	11	46–184	120.6	50.8
Ermelo Pyroclastics[d], mainly central Rocky Creek Block 321.3 ± 3.2 Ma										
14	Ermelo Pyroclastics[d]	AIEA	30.0198	150.3509	lithic ash-flow tuff	80	10	96–235	79.1	29.5
15	Ermelo Pyroclastics[d]	AIEB	30.0198	150.3509	lithic ash-flow tuff	180	10	66–157	79.1	29.5
16	Ermelo Pyroclastics[d]	AIEC	30.0270	150.3623	ash-flow tuff	~50	10	113–188	237.8	27.3
17	Ermelo Pyroclastics[d]	AIED	30.0270	150.3623	ash-flow tuff	~50	10	110–448	237.8	27.3
18	Ermelo Pyroclastics[d?]	AIYD	29.8063	150.2267	ash-flow tuff?	2 m	10	46–103	265.6	45.0
19	Ermelo Pyroclastics[d]	CIBH	30.0227	150.3473	lithic ash-flow tuff	100	12	4.0–285	91.7	45.0
20	Ermelo Pyroclastics[d]	CIBI	30.0278	150.3600	ash-flow tuff	290	12	76–824	241.4	42.2
21	Ermelo Pyroclastics[d]	CIBJ	30.0353	150.3427	ash-flow tuff	300	12	30–151	242.1	39.3
Clifden Formation, Rocky Creek – Boomi Creek region										
22	Tuff ~50 m below Glen Idle Rhyolite?[a,d]	CIBF	30.1187	150.2805	dark dense tuff?	25	12	757–2456	85.6	33.7
23	Siliciclastic ~40 m + CIBF	CIBE	30.1190	150.2822	siliciclastic	~25	11	0.2–2.7	85.6	33.7
24	Tuff ~5 m – Peri Rhyolite[a,d]	AIPA	30.0459	150.2930	ash-fall? tuff	~30	11	0.6–1.0	79.0	26.3
25	tuffaceous ~10 m – AIPB	CICF	30.1928	150.2899	volcaniclastic siltstone	70	12	1.4–5.0	73.5	20.8
26	Peri Rhyolite[a,d] 318.0 ± 3.4 Ma	AIPB	30.1928	150.2899	buff, phenocryst ash-fall tuff	110	10	0.1–1.2	73.5	20.8
27	Peri Rhyolite[a,d] 318.0 ± 3.4 Ma	CIBD	30.1187	150.2860	buff, phenocryst ash-fall tuff	120	12	0.1–1.7	77.4	26.4
28	Tuffaceous ~125 m + AIPB	CICE	30.1931	150.2931	silicified volcaniclastic ss.	425	10	1.6–3.9	94.8	25.4
Clifden Formation, Caroda Formation?, Boomi Creek region										
29	Kooringal Dacite[d]	CICA1-5 6–12	30.1961	150.2789	ash-flow tuff	700	12	12.2–359.0 1.1–4.2	243.9	27.3
30	Boomi Rhyolite[a,d] ~ Peri Rhyolite	CIBU	30.1967	150.2775	ochre, phenocryst tuff	145	12	0.3–1.2	241.4	22.9
31	Tuffaceous ~15 m + CIBU	CIBV	30.1962	150.2765	vitric tuffaceous beds	90	14	0.6–2.1	241.4	22.9
32	Tuffaceous ~15 m + CIBV	CICB	30.1961	150.2758	silicic volcaniclastic ss.	25	12	1.1–3.6	236.3	25.4
33	Tuffaceous ~15 m + CICB	CICC	30.1958	150.2750	silicic volcaniclastic ss.	75	12	0.5–3.6	236.3	25.4
Rocky Creek Conglomerate, Nandewar region										
34	Plagyan Ignimbrite[d]	AILB	30.3840	150.3385	dark phenocryst tuff	14 m	10	61–11 140	68.2	28.8
35	Plagyan Ignimbrite[d]	AILC	30.3840	150.3385	red, phenocryst tuff	280	10	412–982	68.2	28.8

Table 1 (Cont.)

Site no.	Formation/Member/Flow/Site	Acronym	Lat (°S)	Long (°E)	Lithology	Section (cm)	Sample	Initial intensity (mAm^{-1})	Dip direction (°)	Dip (°)
Rocky Creek Conglomerate, Boomi Creek region										
36	Conglomerate below Mt Amore Pyroclastic Mbr[a] of Plagyan Ignimbrite[d]	CIBT	30.1883	150.3043	conglomerate and volcaniclastic matrix	2.5 m	14	59–1564	93.1	30.6
37	Mt Amore pyroclastic Mbr[a] of Plagyan Ignimbrite[d]	AILA	30.1928	150.3364	red-brown phenocrystic tuff	420	10	45–146	260.9	57.8
Rocky Creek Conglomerate, Rocky Creek region										
38	Lithic tuff	CIBW	30.0448	150.3010	splintery lithic tuff	40	12	5.1–54	74.5	18.0
39	Vitric tuff-diamictite	CIBS	30.0447	150.3010	vitric tuff-diamictite	50	12	3.4–90	74.5	18.0
40	Vitric tuff-diamictite	CIBP	30.0453	150.3043	vitric tuff-diamictite	~30	12	62–1405	76.4	17.5
Lark Hill Formation, Rocky Creek region										
41	Eulowrie Pyroclastics[a,d] 310.6 ± 4.0 Ma	CIBO	30.0423	150.3067	reddish vitric tuff	~50	12	1.6–4.3	83.7	13.0
42	Eulowrie Pyroclastics[a,d] 310.6 ± 4.0 Ma	CIBN	30.0423	150.3078	reddish vitric tuff	30	12	0.3–12	83.7	13.0
43	Eulowrie Pyroclastics[a,d] 310.6 ± 4.0 Ma	AIRD	30.0370	150.3095	reddish-purple sandy tuffite	~50	10	3.1–38	282.7	6.2
44	Eulowrie Pyroclastics[a,d] 310.6 ± 4.0 Ma	AIRC	30.0370	150.3095	greenish vitric tuff	~70	10	1.3–30	282.7	6.2
45	Eulowrie Pyroclastics[a,d] 310.6 ± 4.0 Ma	AIRA	30.0369	150.3095	reddish vitric tuff	280	10	1.0–224	247.9	14.3
46	Eulowrie Pyroclastics[a,d] 310.6 ± 4.0 Ma	AIRB	30.0369	150.3095	reddish vitric, in part sandy, tuff	325	10	3.0–50	247.9	14.3
Lark Hill Formation, Tops Road, north Rocky Creek Block										
47	Tycannah Rhyodacite[c,d]	CIYJ	29.8210	150.2363	brown-green ash-fall tuff	210	12	55–287	343.5	3.3
48	Conglomerate	CIYK	29.8207	150.2453	conglomerate, and matrix	~50	7+9	869–19 712 18–1509	87.3	21.7
49	Tops Ignimbrite[c,d]	CIYL	29.8210	150.2543	green-purple ash-flow tuff	~50	12	1.1–45	81.1	20.2
50	Tops Ignimbrite[c,d]	CIYM	29.8222	150.2537	green-red ash-flow tuff	3 m	12	0.7–75	72.3	21.4
51	Tuffaceous	CIYN	29.8253	150.2703	red-purple tuffaceous	230	13	84–2471	76.9	21.9
52	Tuffaceous	CIYO	29.8253	150.2703	red-purple tuffaceous	6.5 m	12	26–873	76.9	21.9
KATHROSE THRUST										
Clifden Formation, Pine Cliffs Road										
53	Tuff ~60 m below CIBY	CIBX	30.1082	150.2683	vitric ash-fall tuff	7 m	13	50–134	272.2	79.6
54	Pine Cliffs Rhyolite[a,d]	CIBY	30.1077	150.2672	reddish vitric tuff	~30	12	801–5052	272.2	79.6
Rocky Creek Conglomerate, Pine Cliffs Road/Llangollan										
55	Rocky Creek Conglomerate	CIBZ	30.1078	150.2650	multicoloured ash-fall tuff	10 m	15	9–59	257.0	78.4
56	Rocky Creek Conglomerate	CIBQ	30.1472	150.2660	yellowish, reworked? tuff	60	12	17–66	272.3	38.4
DARTHULA THRUST										
Rocky Creek Conglomerate, Boomi Creek region										
57	Quartz ss near Hazelvale Rhyolite[a,d]	CIBL	30.2040	150.2682	red ash-fall tuff	320	12	27–78	255.0	33.2
58	Conglomerate and ss. lenses CIBL 40 m+	CIBM	30.2035	150.2687	conglomerate and ss lenses	~50	9+9	0.3–321 0.5–293	255.0	33.2
59	Pound Creek andesite tuff[a]	CICD	30.2048	150.2645	reddish ash-flow tuff	~50	12	4.4–10.1	270.4	32.5
60	Tuffaceous beds ~20 m + Pound Creek andesite tuff[a]	CIBK	30.2048	150.2645	brownish volcaniclastic ss.	360	12	40–336	273.5	32.5
Rocky Creek Conglomerate, Darthula region										
61	Tuffaceous beds above Pound Creek andesite tuff[a]	CIBC	30.1675	150. 1893	grey-purple tuffaceous	2 m	15	91–254	262.7	25.0
62	Tuff (western)[a], below CIBA	CIBB	30.1902	150. 1835	deep red phenocrystic tuff	~50	12	106–331	261.0	13.4
63	Darthula Rhyodacite[a,d]	CIBA	30.1933	150. 1813	purple ash-flow tuff	2 m	12	26–85	261.0	13.4
Lark Hill Formation, Llangollan										
64	Tuff unit 20[a]	CIBR	30.1465	150.2597	unwelded pyroclastics	2 m	12	4–1670	314.6	11.2

[a]Hocking (1973); [b]Yeates (1970); [c]Johnson (1971); [d]name reserved by J. Roberts (pers. comm. 2000: see Opdyke *et al.* 2000), pending formal definition.

SAMPLING

Sampling strategy

An initial pilot study of the Tamworth Belt succession, on the Lower Carboniferous volcano-sedimentary succession of the Upper Rouchel region, demonstrated that ignimbritic volcanics can retain their primary magnetisation, despite conditions leading to pervasive overprinting of intercalated volcaniclastic rocks. Widespread occurrence of magnetic overprints of Permo-Carboniferous (Schmidt 1988; Klootwijk 1996b) and Permo-Triassic age (Lackie & Schmidt 1993) throughout the Tamworth Belt succession and the Carboniferous succession of the Texas and Coffs Harbour Oroclines (Aubourg *et al.* 1994) has become well established. Therefore, sampling of the Rocky Creek Block concentrated on the frequent volcanic and tuff-rich volcaniclastic intercalations—ash-flow tuffs, ash-fall tuffs, lavas, tuffites—throughout the upper Lower to Upper Carboniferous succession.

Sampling of the middle Visean to Westphalian–Stephanian volcanic-rich succession of the Rocky Creek Block (64 sites, 734 samples) has been carried out in three phases. Reconnaissance sampling (17 sites, 171 samples) was carried out in 1987. Preliminary palaeomagnetic results from this collection were listed in Klootwijk (1996a) and their implications were discussed in an initial synthesis of the Carboniferous palaeomagnetism of the Tamworth Belt (Klootwijk 1996b). Extensive follow-up sampling (41 sites, 492 samples) was carried out in 1995 for comprehensive stratigraphic coverage of the successions and for fold tests. Sampling was complemented in 1999 with a small collection (6 sites, 71 samples) to strengthen the palaeomagnetic record of the Clifden Formation. This sampling also addressed the stratigraphic position of the Boomi Rhyolite – Kooringal Dacite succession, generally considered part of the Caroda Formation, but interpreted herein as part of the Clifden Formation.

Samples were collected with a portable drill, an adapted STIHL auger, using a water jacket for cooling and stainless steel barrels for reduced magnetic contamination. Standard drillcores of 2.5 cm diameter and up to 15 cm length were orientated with an inhouse-developed orientation device with integrated magnetic and solar compasses and a clinometer. Sampling concentrated on stratigraphic sections through regionally representative structures, to try to avoid tectonic complications. In general, bedding control for the volcanic rocks could be determined from intercalated volcaniclastic rocks. This was not possible for some sites of the Ermelo Pyroclastics, but indicative bedding control was obtained from faint planar indicators in the pyroclastics.

Sampled sites

Regional geological detail of the sampled region is shown in Figure 4, based on Stroud (1990a, b), and site-specific detail is listed in Table 1. The stratigraphic position of the sampled sites is indicated in stratigraphic sections (Figure 6) after J. Roberts (*in* Opdyke *et al.* 2000 figures 4, 5).

Sampling in the northern Rocky Creek Block concentrated on volcanic rocks in the Caroda to Lark Hill Formations. Prominent volcanic horizons in the Caroda Formation, that is the Yuendoo Rhyolite and West Lynne Rhyodacite, were sampled mainly in steeply eastward-dipping beds on the fault-duplicated western limb of the Slaughterhouse Syncline (Figure 4) (McKelvey 1968; Yeates 1970; Johnson 1971; Pickett 1994). A single site of the Yuendoo Rhyolite (site 8) was collected on the eastern limb of the syncline for a fold test. Several tuffaceous horizons below the Yuendoo Rhyolite have been sampled on the western (2, 3) and eastern (4) limbs of the syncline. In general, samples from the West Lynne Rhyodacite showed weathering effects associated with the Visean–Namurian hiatus. Site 13, collected on a curved strike ridge, may have been affected by local rotation along a nearby fault. Prominent volcanics in the Lark Hill Formation, that is the Tycannah Rhyodacite and the Tops Ignimbrite, have been sampled along the gently eastward dipping Tops Road section (47–52). Boulders and matrix of a prominent conglomerate level (48) have been sampled for a conglomerate test.

Sampling of the Caroda Formation in the central Rocky Creek Block has been limited to an isolated outcrop of the High Valley andesite tuff (1) in the Back Creek region (Hocking 1973) on the western limb of the Rocky Creek Syncline (Figures 3, 4). In Boomi Creek, directly east of Trevallyn, a volcanic–volcaniclastic succession has been sampled (Kooringal Dacite – Boomi Rhyolite: 29–33), which was mapped as Caroda Formation (Hocking 1973; Stroud 1990a, b; Opdyke *et al.* 2000). Palaeomagnetic and rock magnetic evidence obtained in this study indicates that the succession probably belongs to the Clifden Formation (Figure 6), with the Boomi Rhyolite arguably the equivalent of the Peri Rhyolite, and with the Boomi Rhyolite – Kooringal Dacite succession thus constraining the lower boundary of the Permo-Carboniferous Reverse Superchron.

The Ermelo Pyroclastics (14–21) have been sampled in several fault-duplicated structures on the eastern limb of the Rocky Creek Syncline (McKelvey & White 1964; Stroud 1990b). Bedding control was difficult to establish for some sites (16–18, 20) and had to rely on weakly developed planar structures.

Sampling of the Clifden Formation, Rocky Creek Conglomerate and Lark Hill Formation in the Rocky Creek Syncline has been carried out in well-exposed sections along Rocky Creek (24, 38–46) and Boomi Creek (25, 28–33, 36, 37), with another section through the Clifden Formation (22, 23, 27) sampled in a minor creek descending from the western limb of the Rocky Creek Syncline (Figure 4). A further two Rocky Creek Conglomerate sites have been sampled in the Plagyan Ignimbrite on the southeast flank of the middle Tertiary Mt Kaputar volcano (34, 35) directly beneath the Nandewar flows (White 1965).

Sampling of the Kathrose Thrust Sheet has been limited to five sites. Two Clifden Formation sites (53, 54) have been sampled along the freshly excavated Pine Cliffs Road (Figure 4). A further two sites, one along the same road section (55) and the other near Llangollan (56), were originally mapped as Clifden Formation (Hocking 1973; Stroud 1990b), but were identified as Rocky Creek Conglomerate by J. Roberts (*in* Opdyke *et al.* 2000).

Sampling of the Darthula Thrust Sheet (Figure 4) has been limited to two sections through the Rocky Creek Conglomerate—a lower section in Boomi Creek directly west of Trevallyn (57–60) and an upper section near

Darthula (62, 63)—with a further site in the Lark Hill Formation near Llangollan (64).

MAGNETIC TREATMENT, ANALYSIS AND INTERPRETATION METHODS

Sample preparation

All core samples were sandblasted to remove possible magnetic contamination from the stainless-steel drill barrels, and were sliced into standard specimens, 2.54 cm diameter and 2.2 cm height. Top and bottom of the specimens were ground on a diamond lap to remove shape irregularities and magnetic contamination introduced during slicing. At least two duplicate specimens were prepared from each sample, allowing thermal demagnetisation of one specimen per sample. Alternating-field (AF) demagnetisation was carried out on one representative specimen per site and rock magnetic studies on one other specimen per site.

Palaeomagnetic treatment

One specimen per sample was subjected to progressive thermal demagnetisation in 18–36 steps up to peak temperatures of 695°C. Thermal demagnetisation was carried out in large-volume furnaces (McElhinny *et al.* 1971) at the centre of feedback-controlled Rubens–Helmholtz coil systems (10 coils), which reduce the ambient magnetic field to less than 5 nT over the heating–cooling cycle. To minimise oxidation, an argon atmosphere was maintained during the heating cycle (~1 h), but not during forced cooling in air (~0.5 h). Depending on initial intensity of magnetisation and on intensity decrease during demagnetisation, remanence measurements were carried out on a 2-axis ScT cryogenic magnetometer (resolution 2×10^{-3} mAm^{-1}), a 3-axis 2G-Enterprises cryogenic magnetometer (SRM 760R, 10^{-2} mAm^{-1}), or a Molspin spinner magnetometer (10^{-1} mAm^{-1}). The access ports of the three magnetometers are placed in the centres of 6-coil Helmholtz systems to reduce systematic pick-up of viscous magnetisation during measurement. All magnetometers are connected via a LAN to a central database for online data storage and retrieval. The system facilitates real-time inspection of Zijderveld (1967) plots so that informed decisions can be made during measurement about continuation of the demagnetisation treatment. Magnetochemical changes have been monitored through low-field susceptibility measurements with a Digico bulk-susceptibility bridge after each remanence measurement step. All AF demagnetisations were carried out with the automated AF demagnetiser attached to the 2G-Enterprises magnetometer in 27 steps up to 160 mT. All AF demagnetisations above 25 mT peak value have been subjected to a six-axis cyclic demagnetisation procedure to counteract possible gyromagnetic effects (Giddings *et al.* 1997). After completion of demagnetisation the remanence and susceptibility data have been transferred to an ORACLE database on the Australian Geological Survey Organisation's Solaris SUN Unix system for further processing.

The specimens have been analysed individually for linear directions, planar elements and Hoffman–Day (Hoffman & Day 1978) directions with an interactive graphics analysis program, linked to the ORACLE database and based in part on Kirschvink's (1980) Principal Component Analysis method. A three-pass procedure for directional analysis of multicomponent magnetisations has been followed. The first two passes have been applied only for initial interpretation of demagnetisation data for the pilot specimens of the 1987 collection (approximately 30% of the samples for these 17 sites). The third pass has been applied to the entire sample collection (64 sites). In the first pass, Zijderveld and equal-area plots of demagnetisation data for individual pilot specimens have been scrutinised with the interactive graphics analysis program and relevant linear or Hoffman–Day components that showed a visually acceptable fit to the data have been determined. This determination has been routinely carried out in the geographical reference frame. Based on experience, such a visual analysis of individual specimen results, although by nature somewhat subjective, is to be preferred above batch-type analysis of specimens following more objective, but pre-set and rigid acceptance criteria. In the second pass, density distributions in geographical and stratigraphic coordinates of linear and Hoffman–Day directions determined in the first pass have been scrutinised, per site or per formation, for concentrations and/or distributions that could be interpreted in terms of the Carboniferous pole path for the Tamworth Belt. In the third and final pass all specimens, including the pilot specimens already provisionally interpreted during the first two passes, have been (re-)analysed following the first pass procedure. Site-mean directions have been determined from linear components and Hoffman–Day components for a few specimens for which no linear components could be determined. Finally, pole positions have been determined for individual sites and for sites grouped per formation or per stratigraphic interval and have been interpreted in terms of a meaningful pole path.

Rock magnetic treatment

Various rock magnetic tests have been carried out on one representative specimen per site in order to determine magnetic granulometry and magnetic mineral content. This involved: (i) the Lowrie–Fuller (Lowrie & Fuller 1971) test for an indication of domain structure; (ii) the [(Saturation) isothermal remanent magnetisation (IRM)] acquisition test for magnetic mineral determination; (iii) the three-axis Lowrie (1990) test for identification of magnetically weak minerals in particular; and (iv) low-field, low- and high-temperature susceptibility cycling (Schmidt 1993; Hrouda 1994) for determination of granulometry, identification of magnetic minerals and for an indication of possible chemical alterations during thermal cleaning. The first three tests have been carried out on the same specimen, which has been cleaned prior to each test in the maximum 250 mT field obtainable with the automated AF demagnetiser on the 2G-Enterprises cryogenic magnetometer. The fourth test has been carried out on hand-crushed powders of one other representative specimen per site. The Lowrie–Fuller test showed for all sites, without exception, higher resistance to AF demagnetisation for weak-field anhysteretic remanent magnetisation (ARM) than for IRM. This uniformity conveys very little, other than adding to doubts about

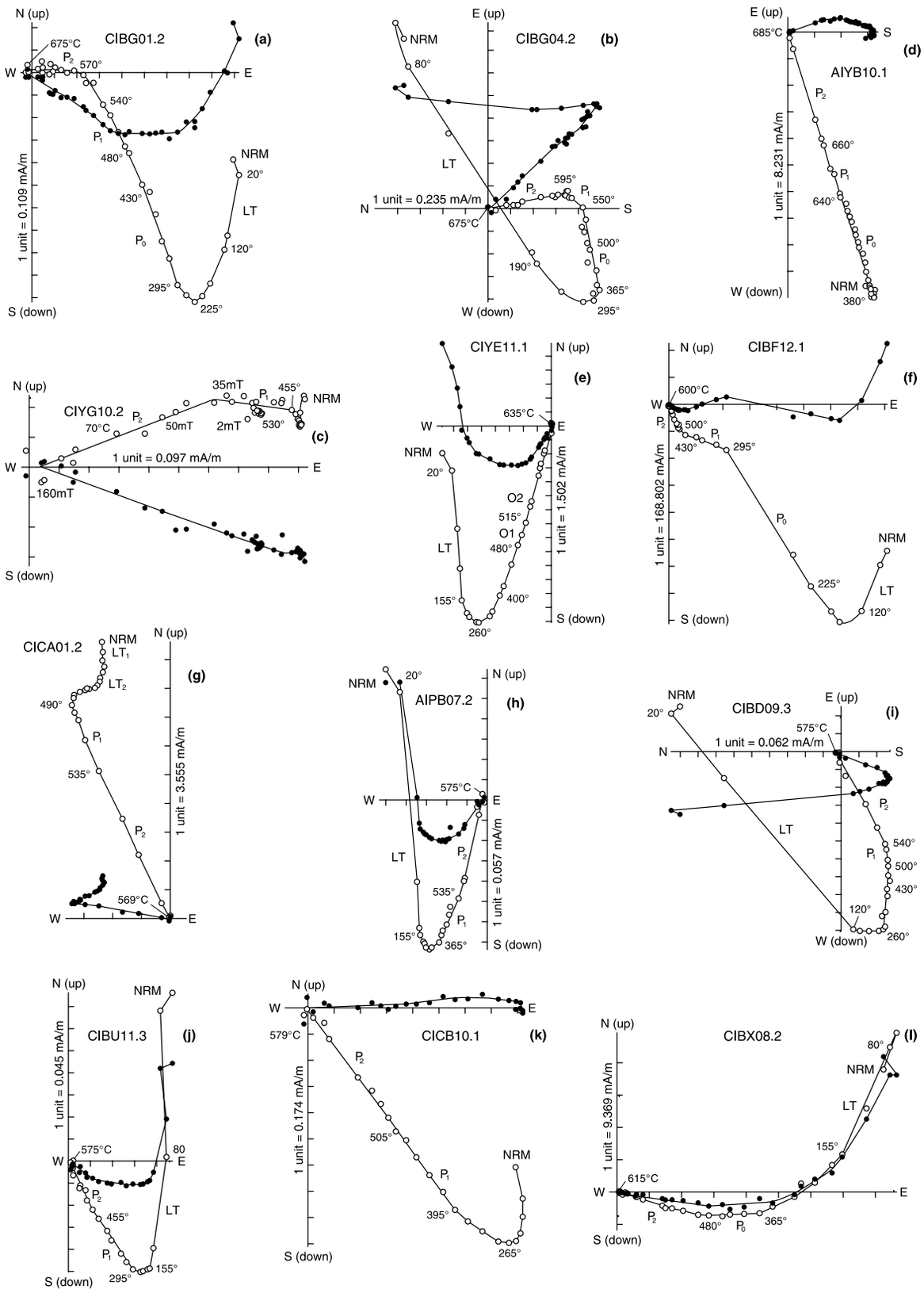
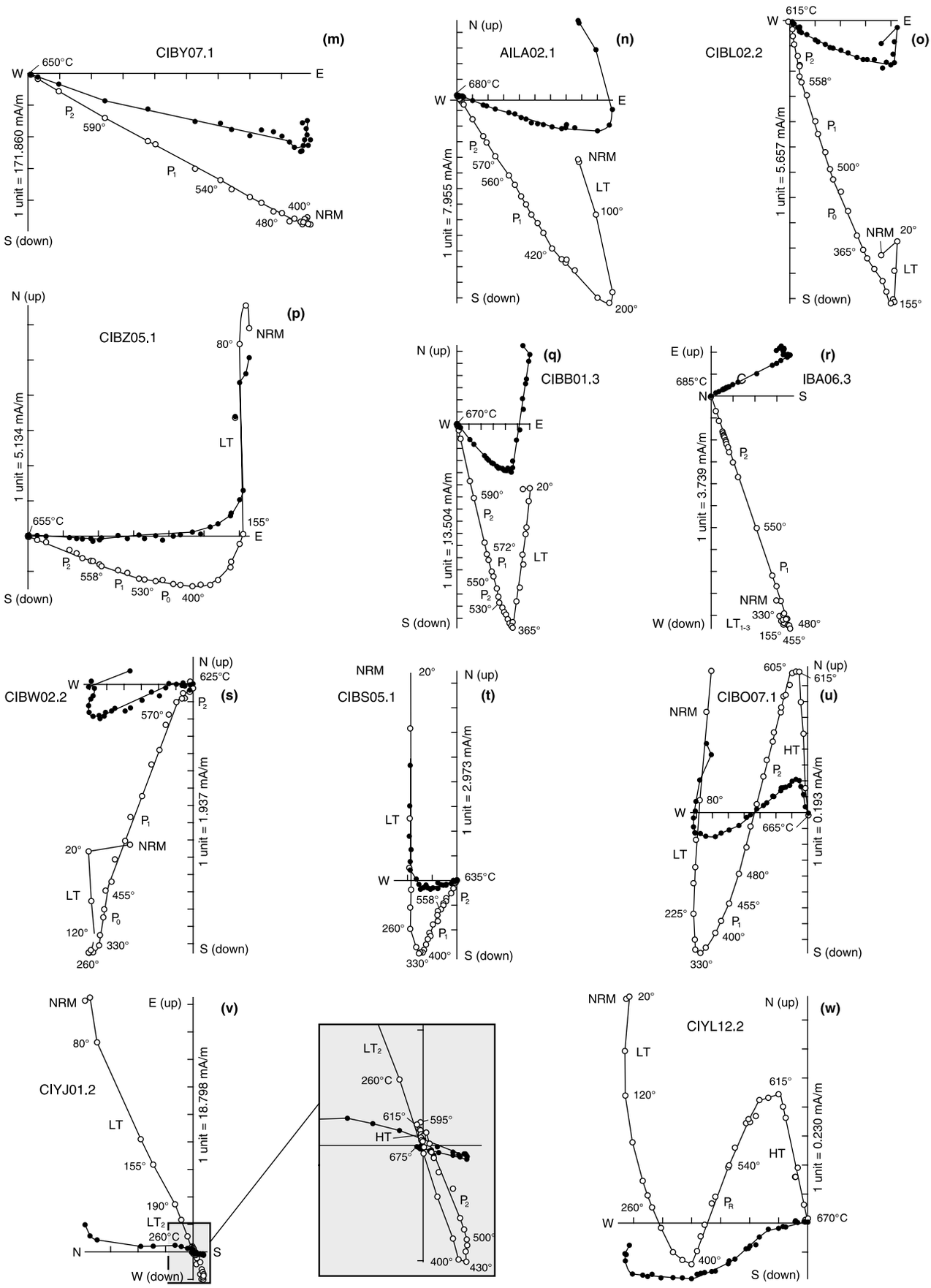


Figure 7 Zijderveld (1967) plots of representative specimens during thermal and alternating-field demagnetisation. The plots are shown in geographical (*in situ*) coordinates for ease of identification of post-folding overprints. The circles indicate successive positions of the end-points of the resultant magnetisation vector during progressive demagnetisation: ○, projections on the vertical east–west or vertical north–south plane; ●, projections on the horizontal plane. Numbers denote successive peak temperatures in °C of the heating cycles, or peak alternating fields in mT. LT_n, low-temperature overprint; Pn, primary component (Table 2) or incompletely cleaned pseudo-components (Table 5*) as discussed in the text and Appendix 1*; shaded box in (v) indicates a partial enlargement.


(m)
CIBY07.1
650°C
W
E
P2
590°
P1
540°
400°
480°
NRM
1 unit = 171.860 mA/m
S (down)
(n)
AILA02.1
N (up)
680°C
W
E
P2
570°
560°
P1
420°
NRM
LT
100°
200°
1 unit = 7.955 mA/m
S (down)
(o)
CIBL02.2
615°C
W
E
P2
558°
P1
500°
P0
NRM
20°
365°
LT
155°
1 unit = 5.657 mA/m
S (down)
(p)
CIBZ05.1
N (up)
NRM
80°
LT
655°C
155°
E
P2
558°
P1
530°
P0
400°
1 unit = 5.134 mA/m
S (down)
(q)
CIBB01.3
N (up)
670°C
W
E
20°
590°
P2
572°
P1
LT
550°
P2
530°
365°
1 unit = 13.504 mA/m
S (down)
(r)
IBA06.3
E (up)
685°C
N
S
P2
550°
P1
NRM
330°
480°
155°
455°
1 unit = 3.739 mA/m
W (down)
LT1-3
(s)
CIBW02.2
N (up)
W
625°C
P2
570°
P1
20°
NRM
LT
455°
P0
120°
330°
260°
1 unit = 1.937 mA/m
S (down)
(t)
CIBS05.1
NRM
20°
N (up)
LT
W
635°C
558°
P2
260°
P1
330°
400°
1 unit = 2.973 mA/m
S (down)
(u)
CIBO07.1
605°
N (up)
615°
NRM
HT
P2
80°
W
665°C
LT
480°
225°
455°
P1
400°
330°
1 unit = 0.193 mA/m
S (down)
(v)
CIYJ01.2
NRM
E (up)
80°
LT
155°
190°
LT2
260°C
N
S
W (down)
1 unit = 18.798 mA/m
LT2
260°C
615°
595°
HT
675°
P2
500°
400°
430°
(w)
CIYL12.2
NRM
20°
N (up)
LT
120°
615°
HT
540°
PR
260°
W
670°C
400°
S (down)
1 unit = 0.230 mA/m

the applicability of the Lowrie–Fuller test (Dunlop & Özdemir 1997), and methodology and results are not further detailed.

IRM ACQUISITION TEST

IRM has been acquired along the z-axis (axial direction) of the samples. For the 1987 and 1995 collections (Table 1, AI and CIY/CIB prefixes) IRM has been acquired with a PYE Scientific Instruments (3′′) electromagnet up to 500 mT and with the Newport Instruments-E (7′′) electromagnet up to 2000 mT, 23 steps in total. For the 1999 collection (Table 1, CIC prefix) IRM has been acquired with the 2G-Enterprises pulse magnetiser in 24 steps up to 2700 mT. Measurements have been carried out on a Molspin magnetometer.

LOWRIE TEST

Using the same electromagnets as for IRM acquisition, the specimens have been magnetised successively along three orthogonal axes in constant fields of 2000 mT (z-axis), 400 mT (y-axis), and 120 mT (x-axis). The specimens have then been thermally demagnetised in up to 31 steps up to 685°C in an argon gas environment. Measurements have been carried out on Molspin and ScT magnetometers.

LOW-FIELD, LOW- AND HIGH-TEMPERATURE SUSCEPTIBILITY CYCLING

Changes in susceptibility during low-temperature (–195°C) and high-temperature (700°C) cycling were monitored with

Table 2 Primary high-temperature (HT) components in stratigraphic coordinates.

Site no.		Direction						South Pole position				
	Comp	Decl (°)	Incl (°)	KD	α95 (°)	*n*	Polarity	Lat (°S)	Long (°E)	Palaeolat (°S)	DP (°)	DM (°)
1	P2	132.5	–13.8	25.4	8.8	12	R	31.3	271.4	7.0N	4.6	9.0
5	P	118.6	30.7	35.4	9.4	8	R	32.6	238.9	16.5	5.9	10.5
6	P2	112.7	28.2	432.3	2.2	11	R	26.8	237.7	15.0	1.3	2.4
(7	*P2* [a]	*137.6*	*29.5*	*52.3*	*7.2*	*9*	*R?*	*48.8*	*2504*	*15.8*	*4.4*	*7.9)*
8	P2	106.8	34.7	23.3	10.2	10	R	23.5	231.0	19.1	6.8	11.7
(9	*P2* [a]	*144.0*	*18.9*	*91.9*	*4.6*	*12*	*R?*	*50.9*	*263.5*	*9.7*	*2.5*	*4.7)*
(14	*P?*	*297.6*	*–51.0*	*49.5*	*6.9*	*10*	*N*	*37.1*	*221.4*	*31.7*	*6.3*	*9.4)*
(15	*P?*	*313.1*	*–46.1*	*36.3*	*15.5*	*4*	*N*	*49.0*	*231.9*	*27.5*	*12.7*	*19.8)*
(19	*P2?*	*103.1*	*68.1*	*32.7*	*7.4*	*13*	*R?*	*30.9*	*195.7*	*51.2*	*10.4*	*12.4)*
(22	*P2*	*101.1*	*33.8*	*30.9*	*7.6*	*13*	*R?*	*18.5*	*229.1*	*18.5*	*4.9*	*8.7)*
24	P2	164.3	78.6	20.1	11.8	9	R	50.8	159.5	68.1	21.1	22.2
26	P2	156.9	71.4	62.0	5.8	11	R	59.4	175.7	56.1	9.0	10.2
27	P2	156.0	66.7	78.9	4.7	13	R	63.6	187.0	49.3	6.4	7.8
(28	*P2?* [b]	*197.0*	*78.1*	*25.6*	*9.2*	*11*	*R*	*51.7*	*139.7*	*67.1*	*16.3*	*17.3)*
29	P2	0.6	–70.6	451.8	2.2	11	N	65.3	149.5	54.8	3.2	3.7
(29	*P2?*	*49.9*	*–62.6*	*79.7*	*4.5*	*14*	*N*	*48.5*	*94.0*	*44.0*	*5.5*	*7.0)*
30	P2	167.3	64.5	35.7	7.8	11	R	71.1	178.1	46.4	10.0	12.4
31	P2	194.2	72.5	27.9	7.4	15	R	60.8	134.7	57.7	11.6	13.1
32	P2	155.8	72.9	83.0	4.8	12	R	57.3	173.7	58.4	7.6	8.5
33	P2	153.0	73.6	30.5	7.6	13	R	55.5	174.3	59.5	12.3	13.7
37	P2	224.3	58.0	131.4	4.2	10	R	52.9	85.5	38.6	4.6	6.2
39	P2	218.0	78.4	14.6	12.4	11	R	46.3	130.5	67.7	22.0	23.3
40	P2	204.6	73.7	29.2	9.7	9	R	56.1	128.2	59.7	15.7	17.5
41	P2	209.1	76.9	137.7	3.6	13	R	50.7	131.4	65.0	6.2	6.6
44	P2	187.2	76.9	271.8	2.8	11	R	54.7	145.1	65.0	4.8	5.2
45	P2	224.9	81.0	170.8	4.3	8	R	41.4	133.9	72.5	7.9	8.2
46	P2	221.1	78.9	41.5	7.2	11	R	44.8	130.6	68.6	12.9	13.6
47	P2	217.5	72.2	87.1	4.5	13	R	52.2	117.8	57.3	7.0	7.9
49	P	237.6	76.3	15.1	12.1	11	R	40.6	121.1	64.0	20.8	22.5
(50	*P2*	*164.5*	*73.0*	*23.9*	*11.6*	*8*	*R?*	*59.4*	*166.2*	*58.6*	*18.4*	*20.6)*
51	P	207.5	72.6	20.8	15.1	6	R	56.1	124.2	57.9	23.7	26.7
52	P	211.6	66.9	26.6	9.0	11	R	59.1	108.7	49.5	12.3	14.9
53	P2 [c]	233.8	74.0	18.0	8.9	16	R	43.6	116.6	60.2	14.5	16.1
54	P2	236.2	66.8	324.4	2.4	12	R	44.0	101.5	49.3	3.3	4.0
55	P2	188.8	75.2	21.6	16.8	5	R	57.4	142.6	62.2	28.2	30.8
57	P2	225.9	64.3	432.5	2.0	13	R	51.3	97.6	46.1	2.6	3.2
62	P2	177.4	74.9	560.9	1.7	14	R	58.5	152.5	61.7	2.8	3.1
63	P2	192.3	67.5	1081	1.2	14	R	67.8	129.1	50.4	1.7	2.0

Data of ambiguous primary or secondary origin are shown in italics and between parentheses (see discussion in text and Appendix 1*).
[a]Incomplete separation of LT component?; [b]Early Permian overprint?; [c]uncertain interpretation.
K (KD, KP), best estimate of the precision parameter κ (Fisher 1953); DP, DM, half-angles of the oval of 95% confidence around the mean pole.
n = number of specimens.

Table 3 Fold test (McFadden 1990), primary components (HT).

Site				*In situ*					Unfolded				Confidence
Result no.[a]	Formation/Member	*n*	Site Nos (Comp)[b]	Dec	Inc	K	A95	SCOS2[c]	Dec	Inc	K	A95	SCOS2[c]	level (%)[d]
1	Yuendoo Rhyolite	3	5(P), 6, 8	136.5	47.0	1.8	156.4	2.981	112.6	31.0	189.0	9.0	0.747	99
2	Peri Rhyolite	3	26, 27, 30	182.6	69.0	12.9	35.9	2.966	160.4	69.0	568.8	5.2	2.131	–
3	Upr Clifden Fm.– lwr Rocky Creek Congl.[e]	3	37, 54, 57	112.4	52.6	12.4	36.7	1.707	228.2	63.1	231.3	8.1	1.943	–
4	Rocky Creek Conglomerate	5	39, 40, 55, 62, 63	156.3	72.1	4.5	40.7	2.882	195.0	74.3	210.5	5.3	1.788	95
5	Lark Hill Fm. (Tops Road)	4	47(P2), 49, 51, 52(P)	234.0	58.1	45.2	13.8	3.462	217.0	72.3	224.7	6.1	1.038	99
6	Lark Hill Fm. (Rocky Creek)	4	41,44–46	182.9	82.1	32.8	16.3	3.626	219.0	80.5	159.4	7.3	0.627	99
7	Lark Hill Fm. combined	8	41, 44–47, 49, 51, 52	224.1	71.2	18.3	13.3	7.399	217.7	76.4	132.7	4.8	1.114	99

The McFadden (1990) fold test has been applied routinely. The McFadden (1998) fold test requires at least two observations per cluster, a requirement not met by all datasets.
[a]see Figure 14.
[b]Component P2 unless indicated otherwise.
[c]Critical values of the test statistic SCOS2 for the correlation test (McFadden 1990) at the 95% (99%) confidence limits are for number of sites: 3, 2.076 (2.662); 4, 2.335 (3.180); 5, 2.609 (3.573); 8, 3.298 (4.562).
[d]Indicates the fold tests that show at the 95% (99%) confidence level correlation between between magnetisation and bedding in the *in situ* position and no correlation in the unfolded position.
[e]Questionable primary origin of magnetisation.
K (KD, KP), best estimate of the precision parameter κ (Fisher 1953).

an AGICO CS-3 bridge (Hrouda 1994). The high-temperature cycle was carried out in argon gas to reduce oxidation.

Magnetic fabric determination

Anisotropy of magnetic susceptibility was determined with the AGICO KLY-3S slow-spinning magnetometer on magnetically untreated specimens, one per sample and up to 10 or 12 specimens per site. The KLY-3S operates at 875 Hz, with a sensitivity of 3×10^{-8} SI for bulk measurements and 2×10^{-8} SI for anisotropy measurements. Mean-tensor coefficients, eigenvalues and error ellipsoids were determined with AGICO's ASTA program and also with the ANIPO program developed at the Fort Hoofddijk palaeomagnetic laboratory, University of Utrecht, the Netherlands. Both programs apply Jelinek's (1978) tensorial statistics to determine site-mean anisotropy of magnetic susceptibility (AMS) results, with more flexible data editing for ANIPO data files (ASCII) versus ASTA's (binary).

PALAEOMAGNETIC RESULTS

General component structure

Thermal demagnetisation generally resulted in better separation of multicomponent magnetisations than AF demagnetisation. Therefore, thermal demagnetisation was routinely carried out on at least one specimen per sample. In general, AF demagnetisation has been limited to one specimen per site with the results used as a first indication of the site's magnetisation content and magnetic behaviour. All demagnetisations have been carried out as progressive treatments, allowing full directional analysis of multicomponent magnetisations. An overview of the component structure follows and representative Zijderveld plots are presented in Figure 7. Magnetic components are described in more detail in Appendix 1*.

First-pass inspection of thermal demagnetisation results indicated the general presence of two thermally distinct groups of magnetic components, unblocked at low temperatures (LT) and at higher to high temperatures (HT). Further scrutiny identified a third group of magnetic components, generally with intermediate unblocking temperatures (IT) and identified these as overprint magnetisations. A few sites showed overprints with high to very high unblocking temperatures; for convenience these have been included in the intermediate group.

The LT components are generally removed below 300°C and below 30 mT. They are nearly exclusively of normal polarity with *in situ* directions (mean-site direction: D = 5.6°, I = –64.5°, n = 54, k = 96.0, α_{95} = 2.0°; mean-site pole: 138.3°E, 72.6°S, α_{95} = 2.9°) closer to the present field direction in the sampled region (D = 11°, I = –60.5°) than to the expected direction according to the axial geocentric dipole field model (D = 0°, I = –49°). The HT components are generally removed between 250° and 680°C and 15–160 mT, and are predominantly of reverse polarity with downward and broadly southward directions. Demagnetisation at higher temperatures (upper magnetite and hematite unblocking temperature ranges) mainly shows declination rather than inclination shifts, with appearance of incompletely cleaned

pseudo-components prior to isolation of better cleaned components at the higher temperature steps. Considerable effort has been spent in detailing and identifying pseudo-components and proper components through detailed, prolonged demagnetisation and statistical analysis. This demonstrated that the components with the best statistical parameters (α_{95}), the preferred primary component representatives, were often obtained from the remaining last few percentages of the magnetic signal (Tables 2, 3). Discontinuation of the demagnetisation procedure at lower temperatures would have prevented such distinction, risking determining mean results from less well-cleaned pseudo-components (Tables 5*, 6*). The third group of IT components includes various magnetically softer components and, for convenience, a magnetically very hard component. *In situ* or bedding-corrected directions (Table 7*) suggest that these IT components are indicative of phases of magnetic overprinting of mainly Early Permian, Permo-Triassic (bedding corrected) and Late Cretaceous (*in situ*) age, but also of middle Carboniferous (bedding corrected), Miocene and Holocene age (*in situ*). These IT components are generally unblocked between 100°C and 480°C and below 20 mT, with the exception of the very hard component in the Eulowrie Pyroclastics of the Lark Hill Formation, which is removed between 570°C and 670°C and between 100 mT and 160 mT.

ROCK MAGNETIC RESULTS

IRM acquisition

The IRM acquisition curves show that the magnetisation of most of the volcanic and volcaniclastic rocks is made up of a magnetite phase, generally saturated in fields of 200–300 mT [e.g. Yuendoo Rhyolite, West Lynne Rhyodacite, Ermelo Pyroclastics, Glen Idle Rhyolite, Kooringal Dacite, Plagyan Ignimbrite, Rocky Creek Conglomerate at Rocky Creek, part of the Eulowrie Pyroclastics, Tycannah Rhyodacite, part of the Tops Ignimbrite, Pine Cliffs Rhyolite (Figure 8b), and tuff below the Darthula Rhyodacite] and sometimes in higher fields [part of the Yuendoo Rhyolite (Figure 8a)]. A fair number of volcanics, however, show a distinct high coercivity, hematite or goethite phase, after saturation of a magnetite phase at 200–400 mT [e.g. the High Valley andesite tuff (Figure 8c), part of the Yuendoo Rhyolite, Peri Rhyolite, Boomi Rhyolite (Figure 8d), part of the Eulowrie Pyroclastics, part of the Tops Ignimbrite, and Darthula Rhyodacite].

Three-axis Lowrie test

A minority of the volcanic rocks shows a distinct goethite component, obviously weathering related [e.g. West Lynne Rhyodacite, Ermelo Pyroclastics, Peri Rhyolite (Figure 8f), Boomi Rhyolite, part of the Rocky Creek Conglomerate and part of the Tops Ignimbrite]. The volcaniclastic rocks, being more prone to weathering, show more consistent presence of a goethite component (e.g. the tuffaceous beds near the Hazelvale Rhyolite and above the Pound Creek andesite tuff). Most of the volcanic and volcaniclastic rocks show a distinct magnetite unblocking temperature range [e.g. Yuendoo Rhyolite (Figure 8g), West Lynne Rhyodacite, Ermelo Pyroclastics, Kooringal Dacite, Peri Rhyolite (Figure 8f), Boomi Rhyolite, Plagyan Ignimbrite, Rocky Creek Conglomerate, part of the Eulowrie Pyroclastics, Tycannah Rhyodacite, Tops Ignimbrite, Pine Cliffs Rhyolite and most of the volcaniclastic rocks]. A small number of volcanic rocks show a distinct hematite Curie temperature up to 675°C [e.g. the High Valley andesite tuff (Figures 8e, 9c), part of the Yuendoo Rhyolite (Figures 8h, 9e), the Pound Creek andesite tuff, main part of the Eulowrie Pyroclastics, main part of the Tops Ignimbrite and the Darthula Rhyodacite], additional to magnetite. The Ermelo Pyroclastics and part of the volcaniclastic rocks show distributed unblocking temperature ranges up to 300°C. None of the three-axis Lowrie test results clearly indicate the presence of maghemite or iron sulfides.

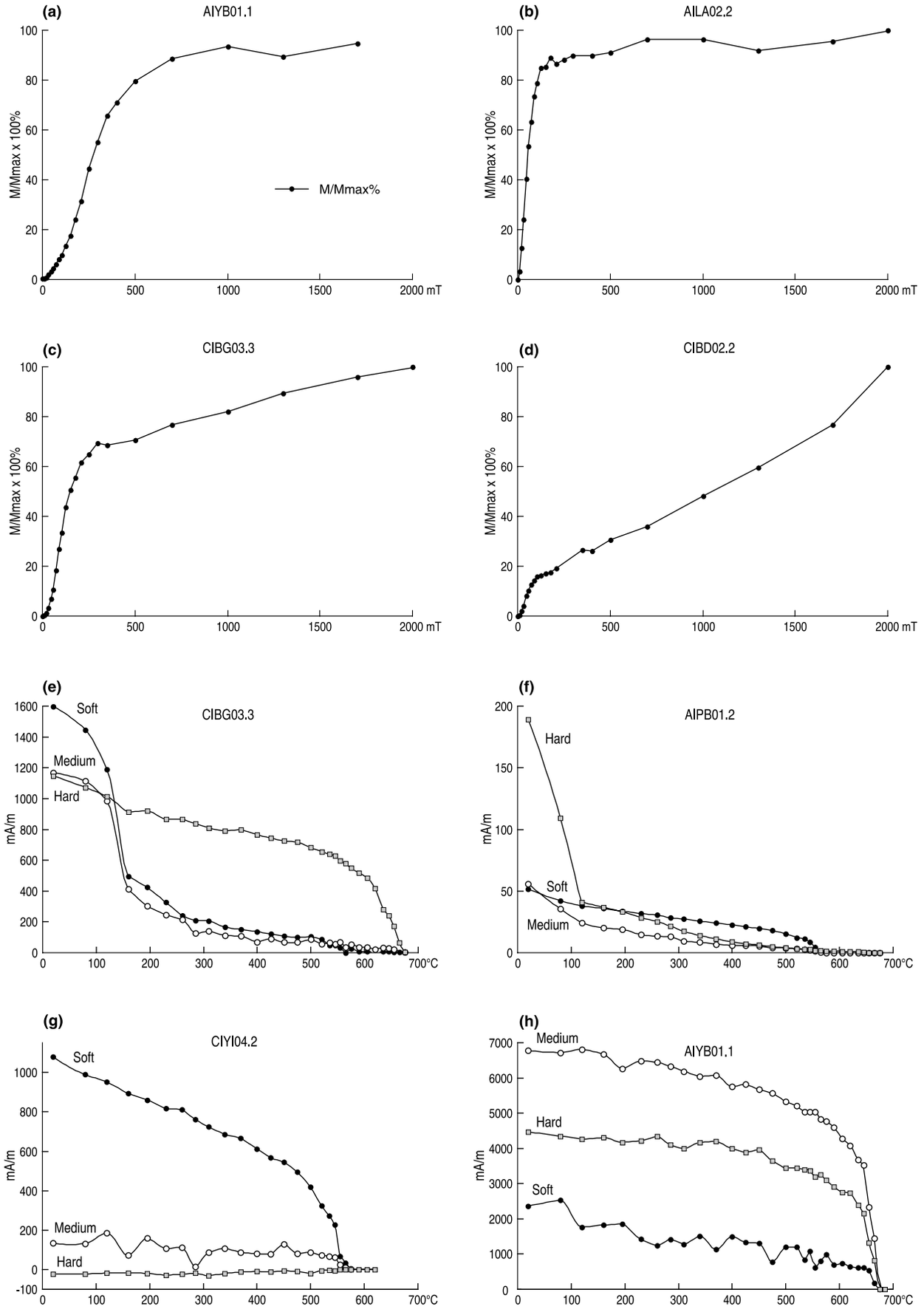
Low-field, low- and high-temperature susceptibility cycling

LOW-TEMPERATURE SUSCEPTIBILITY CYCLING

Low-temperature susceptibility cycling shows the typical, unresolved signature of the multidomain (MD) magnetite isotropic point at approximately –140°C and the Verwey Transition at approximately –150°C (Schmidt 1993) in various volcanic rocks [some of the Yuendoo Rhyolite sites with suspected incomplete separation of an HT component and a present field-like component (or an aberrant HT component direction), West Lynne Rhyodacite, Ermelo Pyroclastics (Figure 9a), a Peri Rhyolite site, the remagnetised Plagyan Ignimbrite sites at Mt Kaputar, the vitric tuff–diamictite succession of the Rocky Creek Conglomerate (Figure 9b), part of the Eulowrie Pyroclastics of the Lark Hill Formation, tuffs below and above the Pine Cliffs Rhyolite and the tuff directly below the Darthula Rhyodacite]. However, the majority of volcanic rocks with well-defined HT components show no or very limited presence of MD magnetite (e.g. the High Valley andesite tuff, most of the Yuendoo Rhyolite, Kooringal Dacite, most of the Peri Rhyolite, Boomi Rhyolite, the Tycannah Rhyodacite, the Tops Ignimbrite, the Pine Cliffs Rhyolite and the Darthula Rhyodacite). The MD magnetite signature shows up more consistently in the volcaniclastic rocks (e.g. in the matrix of the conglomerate horizon in the Lark Hill Formation from the Tops Road section).

There is no clear indication in any of the sites for the Morin Transition at approximately –20°C, but this may merely reflect sensitivity limitations of the CS-3 unit in analysing powders that have not been magnetically concentrated. Some of the volcanic rocks show minor, but distinct, susceptibility peaks between approximately –100°C and –50°C of unclear origin.

Figure 8 (a–d) Examples of isothermal remanent magnetisation (IRM) acquisition in constant fields up to 2000 mT. (e–h) Examples of thermal demagnetisation of IRM three-axis magnetisations following Lowrie (1990). Grey-shaded squares, hard IRM acquired in a 2000 mT constant field; ○, medium IRM acquired at 400 mT; ●, soft IRM acquired at 120 mT.


(a)
AIYB01.1
M/Mmax x 100%
M/Mmax%
2000 mT
(b)
AILA02.2
(c)
CIBG03.3
(d)
CIBD02.2
(e)
CIBG03.3
Soft
Medium
Hard
mA/m
700°C
(f)
AIPB01.2
(g)
CIYI04.2
(h)
AIYB01.1

HIGH-TEMPERATURE SUSCEPTIBILITY CYCLING

High-temperature susceptibility cycling shows the predominant presence of a magnetite component with Curie temperatures up to 580°C [e.g. for the High Valley andesite tuff (Figure 9c), the Ermelo Pyroclastics, Kooringal Dacite, Peri Rhyolite, Plagyan Ignimbrite and vitric tuffs of the Rocky Creek Conglomerate (Figure 9d)]. This magnetite component is often accompanied by a hematite component with Curie temperatures up to 680–705°C (e.g. part of the Yuendoo Rhyolite, West Lynne Rhyodacite, Eulowrie Pyroclastics, Tycannah Rhyodacite, Tops Ignimbrite, Pine Cliffs Rhyolite and Darthula Rhyodacite). The characteristic changeover (lepidocrocite)–maghemite–magnetite (Schmidt 1993) is visible in some of the volcanic rocks [e.g. the Plagyan Ignimbrite (Figure 9f), the tuff below the Glen Idle Rhyolite and the tuff below the Darthula Rhyodacite]. This is also a common observation in the volcaniclastic rocks. The volcanic rocks quite commonly show formation of a substantial magnetite component and less frequently a possible maghemite component after heating to the maximum temperature of ~700°C, indicating ineffectiveness of the argon environment in preventing magnetic phase changes.

MAGNETIC FABRIC ANALYSIS

Measurement results

The volcanic rocks show a low anisotropy of magnetic susceptibility (AMS). Anisotropy values (P) vary generally between 1% and 3% (a normal range for ignimbrites and air-fall tuffs: MacDonald & Palmer 1990) and occasionally up to 5%. The volcaniclastic rocks generally show P values between 1% and 4% with the exception of a 9% value for the matrix of a conglomerate in the Tops Road section (Site 58). The volcanic and volcaniclastic rocks show predominance of oblate over prolate AMS ellipsoid shapes. Well-defined triaxial ellipsoids are also prominent, but various sites do not have a clear magnetic fabric pattern.

The potential of magnetic fabric analysis for determination of transport directions in ignimbrites and transported air-fall tuffs has been well established from pioneering studies on felsic volcanic rocks, mainly Cenozoic, in southwestern North America (MacDonald & Palmer 1990; Palmer *et al.* 1991). Such studies have shown that the K_{min} axis of the AMS ellipsoid generally represents the pole to the foliation or flow-banding plane, that the K_{max} axis parallels the flow- or transport-axis, and that imbrication of coarser components within the volcanic rocks can be used as an indication of transport direction with the source expected in the downdip direction of imbrication. The method can only be applied reliably if it can be established that the AMS ellipsoids reflect primary depositional patterns and not tectonic deformation.

The *in situ* K_{max} axes of the volcanic and volcaniclastic rocks of the Rocky Creek Block show an overall north–south alignment (Figure 10b). The regional tectonic grain is also north–south (Figures 3, 4) and a tectonic origin of the AMS pattern might, therefore, be surmised at first glance. However, unfolding of the principal susceptibility axes according to the attitude of the beds shows that the magnetic foliation planes (K_{max}–K_{int}) essentially agree with the palaeohorizontal (Figure 10a) and, importantly, shows a substantially improved grouping for the K_{min} axes (Figure 10c, d). This strongly indicates that the AMS

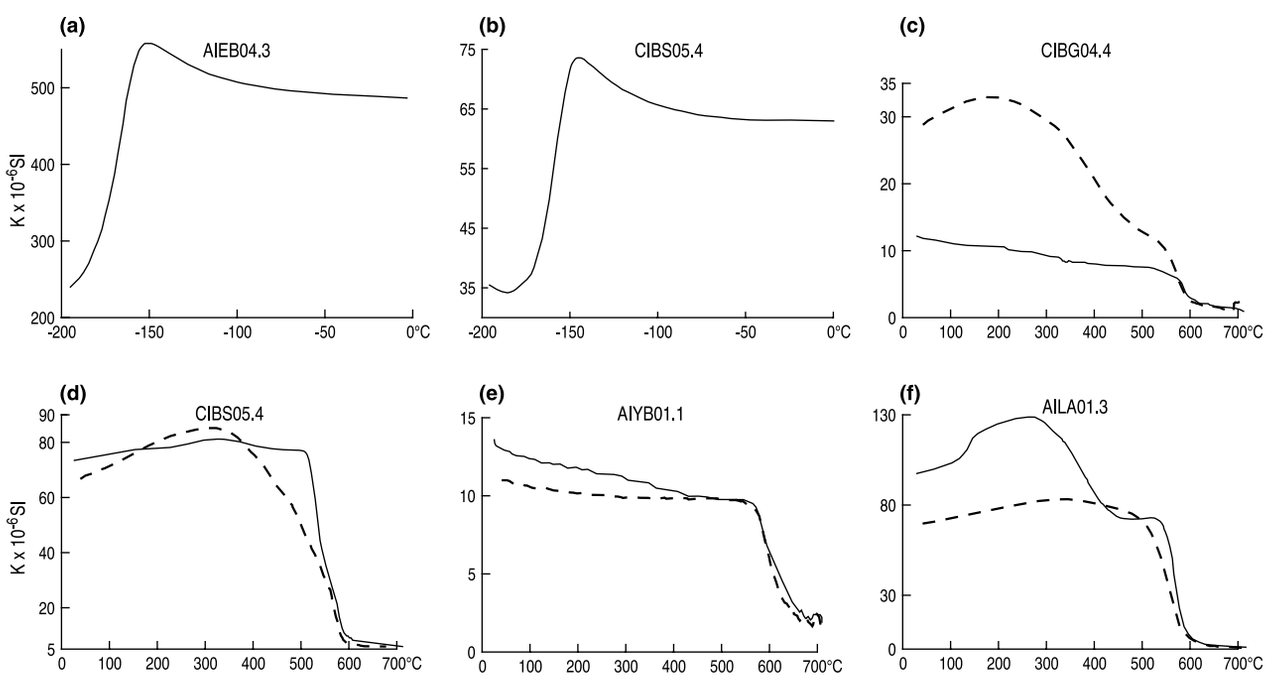


Figure 9 (a, b) Examples of thermomagnetic curves obtained with the AGICO CS-3 cryostat showing changes in susceptibility during warming up from liquid nitrogen temperature. (c–f) Examples of thermomagnetic curves obtained with the AGICO CS-3 furnace showing changes in susceptibility during heating (continuous curve) and cooling (dashed curve) carried out in an argon gas environment.

patterns are primary. Formational consistency in K_{max} directional patterns and variation through the stratigraphy (Figure 11) supports this. The Caroda and Clifden Formations show a gross north–south alignment, whereas the Ermelo Pyroclastics show northwest–southeast alignment (Figure 11a). The Rocky Creek Conglomerate, in particular, and less so the Lark Hill Formation, in contrast show variable patterns probably reflecting their higher energy environment (Figure 11b).

The unfolded magnetic foliation planes of the volcanic units show an about-even distribution of northern and southern dips throughout the succession (Figure 10e). Thus, there is no clear indication from magnetic fabric evidence for regionally or temporally persistent imbrication effects, and no definitive conclusion can be drawn from the AMS results alone on the polarity of transport.

Transport patterns

Regional facies variations, such as thickness of beds, palaeocurrents and proximal versus distal characteristics, indicate a western, albeit unidentified, source region for the volcanic rocks (Leitch 1974; Veevers *et al.* 1982; McPhie 1987; Roberts & Engel 1987; Scheibner & Basden 1998; Wang & Roberts 1999). This field evidence makes up for the lack of AMS control on polarity of transport in the limited number of volcanic sites with broadly east–west-aligned K_{max} axes, pointing clearly to eastward transport directions. However, the Caroda and Clifden Formations show north–south-trending K_{max} axes patterns for which west-to-east facies variations are not discriminatory regarding polarity of transport.

The AMS transport indicators are not conducive to locating a source region for other than, perhaps, the Ermelo Pyroclastics. The general north–south trend of transport directions, in the Caroda and Clifden Formations in particular, suggests distal rather than proximal source regions and may reflect the common evolution of transport directions of further travelled material from following dip to become aligned with topographic grain.

Consistency in transport directions per formation across the three thrust sheets supports the above established general agreement in palaeomagnetic directions between the thrust sheets. Consistent and similar north-northwest–south-southeast transport directions for the Peri Rhyolite and Boomi Rhyolite further substantiates

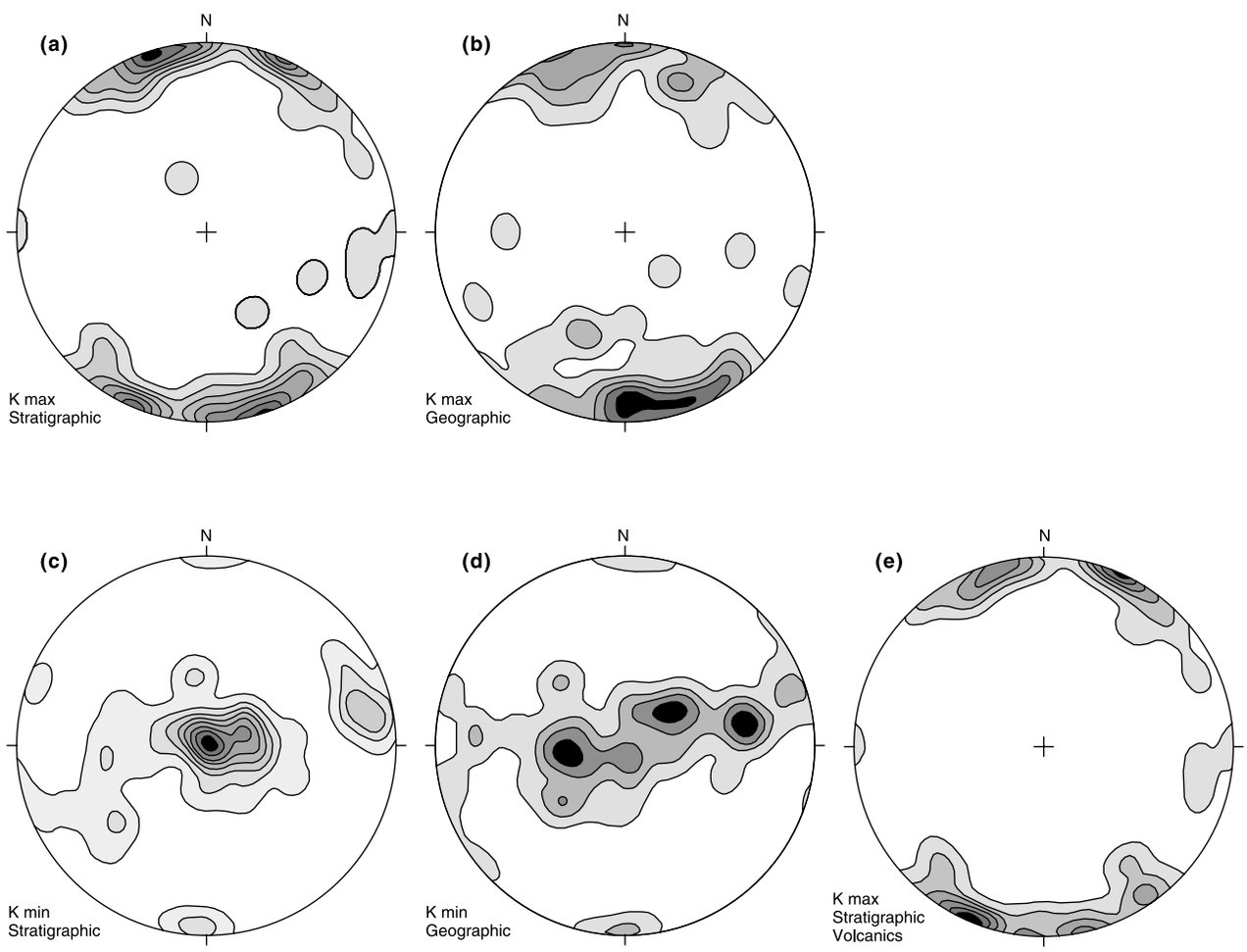


Figure 10 Contoured density distributions of AMS mean-tensorial axes for all measured sites. Lower hemisphere projection; contours are made for each integer value. (a) K_{max} axes, all sites in stratigraphic coordinates. (b) Same in geographical coordinates. (c) K_{min} axes, in stratigraphic coordinates. (d) Same in geographical coordinates. (e) K_{max} axes of volcanic sites in stratigraphic coordinates, no clear evidence for imbrication.

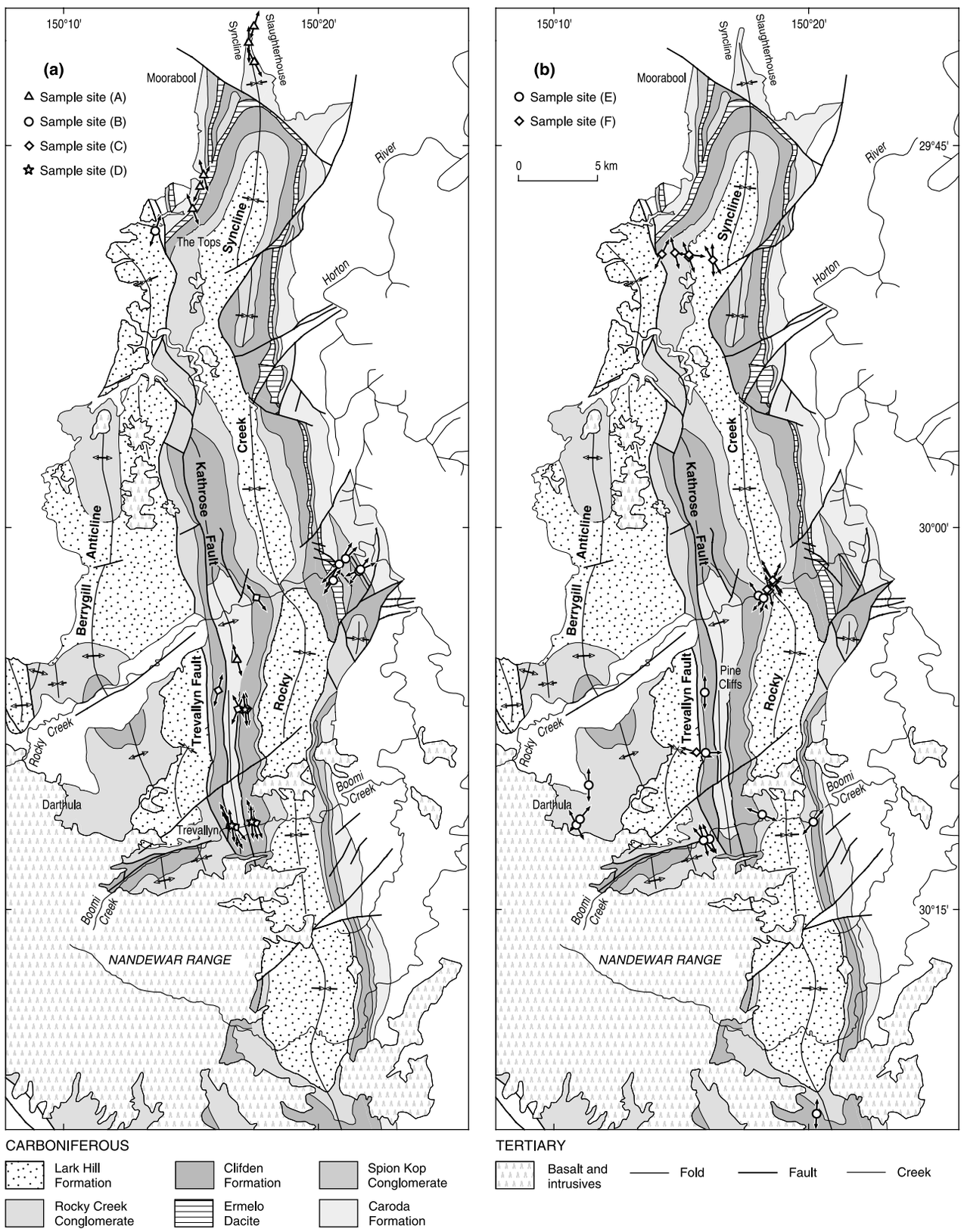


Figure 11 Orientation of AMS mean-tensorial K_{max} axes for individual sites as an indication of transport direction. Note that the polarity of transport direction has not been established in absence of magnetic evidence for imbrication (see Figure 10e). (a) A, Caroda Formation – High Valley andesite tuff, Yuendoo Rhyolite, West Lynne Rhyodacite; B, Ermelo Pyroclastics; C, Clifden Formation from Rocky Creek and Kathrose Thrust Sheets; D, Peri Rhyolite, Boomi Rhyolite. (b) E, Rocky Creek Conglomerate from Rocky Creek, Kathrose and Darthula Thrust Sheets; F, Lark Hill Formation from the Rocky Creek and Kathrose Thrust Sheets.

the above arguments that the two units are correlatable (Figure 11a–d).

INTERPRETATION

Overprint magnetisations: post-folding

MIDDLE–LATE CENOZOIC OVERPRINTS

A magnetically soft overprint of exclusively normal polarity and with *in situ* directions close to the present local field direction is consistently present throughout the sites (Figure 12a, b). This component is unblocked at temperatures up to 300°C and in alternating fields up to 30 mT. Its mean direction is indistinguishable from a present field component in the southward adjacent Werrie Block. Both are similar to an overprint observed on the south coast of New South Wales and interpreted as Early Miocene in age (Nott *et al.* 1991). This set of overprints may be indicative of a regional mid-Tertiary weathering event, although the overprints' exclusive normal polarity and closeness to the present local field direction alternatively may point to a recent viscous origin. A few sites also show another, magnetically slightly harder, component with *in situ* directions close to the present local field (Table 7*; Figure 12c). These are exclusively of normal polarity, with the exception of the Darthula Rhyodacite (Site 63), which shows a reverse polarity component in addition to a normal polarity component. This slightly harder component is unblocked between 100°C and 500°C. Some sites (e.g. 45 and 46 of the Eulowrie Pyroclastics) show inclinations for this harder component that are steeper than the present local field direction and closer to the expected direction for the Late Cretaceous overprint (Figure 12a). Finally, two sites on the southeastern flank of Mt Kaputar (34, 35) show well-defined IT and HT components with *in situ* directions (Figure 12d) comparable to the reverse polarity direction of the directly overlying Miocene Nandewar Volcanics (Wellman *et al.* 1969).

LATE CRETACEOUS OVERPRINT

The Eulowrie Pyroclastics sites and a few other sites (Table 7*) show a characteristic, well-defined, magnetically very hard, component with *in situ* directions close to the expected direction for the Late Cretaceous overprint of normal polarity (D = 9°, I = –75°), or nudging toward the present local field direction (Figure 12e). This component is generally removed between 580°C and 680°C and obviously resides in secondary hematite. The Late Cretaceous overprint is well established in the Australian margin of the Tasman Sea (Schmidt & Embleton 1981; Klootwijk *et al.* in press b), where it is attributed to a thermal event with subsequent cooling, uplift and erosion prior to initiation of sea-floor spreading. Fission-track and fluid-inclusion data indicate acquisition at approximately 95–90 Ma. The Late Cretaceous normal polarity overprint seems to have a wider regional presence than originally surmised and has now been established from fission-track and palaeomagnetic studies in more inland regions (Schmidt & Embleton 1981; O'Sullivan *et al.* 1999; Klootwijk in press b), probably induced by fluids propagated along major structural zones.

Overprint magnetisations: pre-folding

PERMO-TRIASSIC OVERPRINTS

Many of the IT and softer HT components and a minority of harder HT components, predominantly of reverse polarity, show, after correction for bedding, poles in the southwest Pacific with a wide directional coverage, mainly to the north, east and south of New Zealand (Table 7*; Figure 13a). This coverage broadly resembles the outline of the major Permo-Triassic loop on the Australian cratonic pole path (Figure 13b) (Klootwijk 1996b). Comparison with a New England pole path would be preferable for dating of the components, but this is not available. Post-accretion displacement of the New England Orogen versus the craton, while unresolved in detail, is not expected to notably affect such a first-order comparison with the cratonic pole path. The Australian Permo-Triassic loop is ill defined, and its extent is probably under defined. The loop's outline and extent can be judged from the slightly better defined Indian pole path, transferred to Australia within Gondwanaland (Figure 13c) (Klootwijk 1996b). Poles close to the outline of the Australian–Indian Permo-Triassic loop are likely to reflect overprinting associated with the Permo-Triassic Hunter–Bowen Orogeny (Harrington & Korsch 1985b; Veevers *et al.* 1994a). Large-scale west-directed thrusting of the Tamworth Belt across the Bowen–Gunnedah–Sydney Basin system occurred during its Middle Triassic main phase (Korsch *et al.* 1997), facilitating magnetic overprinting through fluid movements along developing fault systems.

EARLY PERMIAN OVERPRINTS

A few components of exclusively reverse polarity, mainly HT components from silicic diamictites and tuffs in the Rocky Creek Conglomerate (Table 7*), show very steep inclinations and poles in general agreement with the Early Permian segment (Figures 12f, 13b, c) of the Australian pole path, which is outlined by only a few indicative poles (e.g. Werrie Volcanics: Schmidt 1988). The main extensional phase of the Bowen–Gunnedah–Sydney Basin system occurred during the latest Carboniferous and Early Permian (Tadros 1993; Veevers *et al.* 1994b; Holcombe *et al.* 1997), followed shortly by oroclinal deformation of the southern New England Orogen. The Meandarra Gravity Ridge (Murray *et al.* 1989), which can be traced for more than 1200 km from the southern Sydney Basin through the Gunnedah Basin (along the western Tamworth Belt) up to the Taroom Trough of the southern Bowen Basin, was emplaced during this extensional phase. Fluid movements associated with this igneous activity may have induced these sporadic HT overprints. It is unclear what has localised these overprints in contrast to the widespread occurrence of Permo-Triassic overprints.

OVERPRINTS VERSUS COMPOSITE COMPONENTS

The Permo-Triassic and Early Permian overprints (Table 7*) could be affected by composite unblocking of a

Carboniferous primary magnetisation and a Holocene overprint, and could thus represent artefacts. The likelihood of this effect has been explored by determining angular distances between overprint directions and planes through bounding component directions. This showed angular distances of ≤5°(≤8°, ≤10°) for 33% (43%, 49%) of

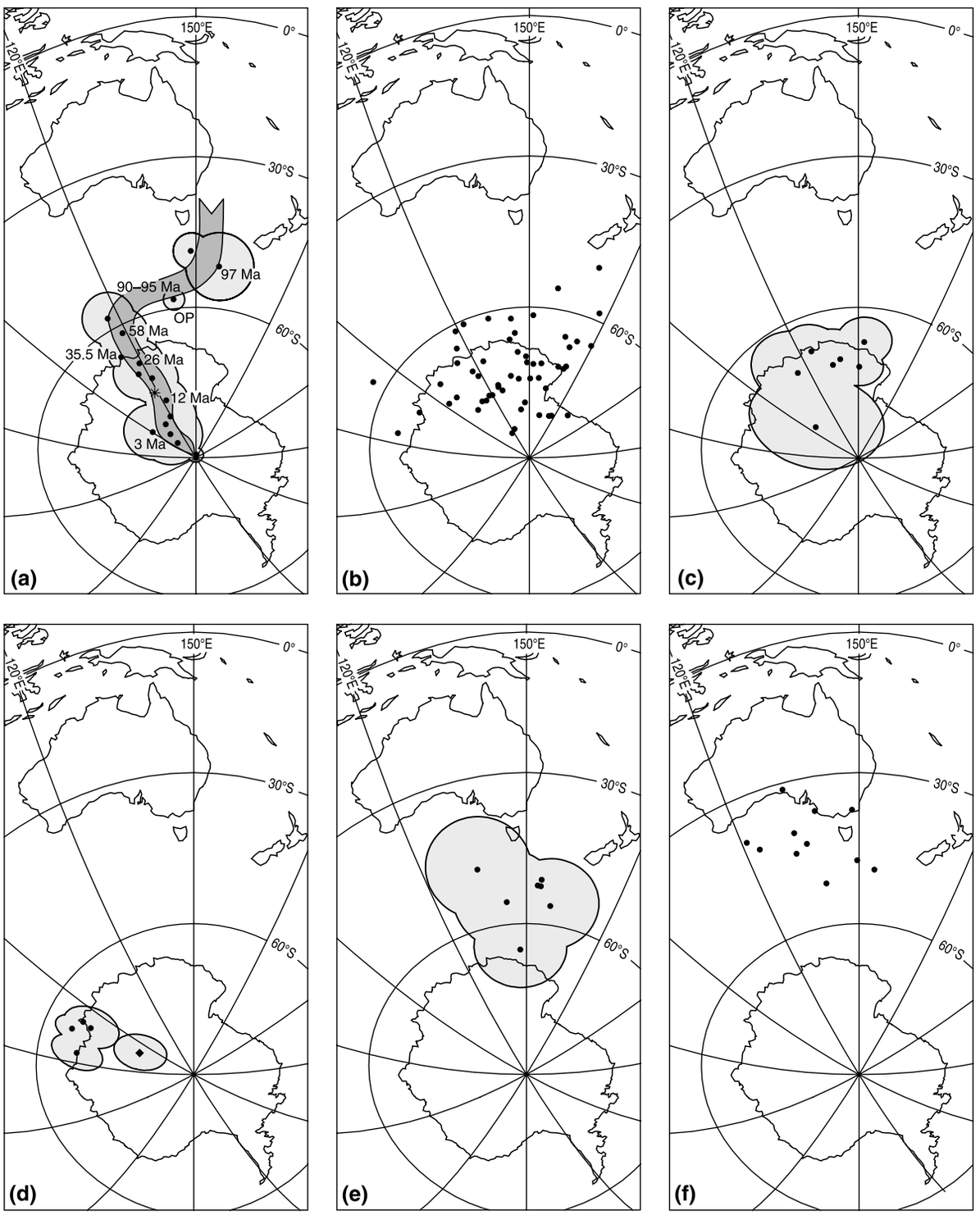


Figure 12 South pole positions for post-folding overprints of Cenozoic (b–d) and Late Cretaceous (e) origin and for pre-folding overprints of Early Permian origin (f). (a) Cenozoic and Late Cretaceous pole path and poles (●) after Idnurm (1994) and Klootwijk (in press a figure 8) for comparison with overprint poles shown in (b–d). ★ indicates the present Virtual Geomagnetic Pole. (b) Wide and prevalent occurrence of 'Recent Field'-like LT overprints (●) of normal polarity that may represent a middle Tertiary weathering event. (c) More restricted occurrence of 'Recent Field'-like IT and HT overprints (●) of normal polarity (Table 7*). (d) Local occurrence of Miocene overprints (●) of reverse polarity related to the Nandewar Volcanics (◆) (Table 7*). (e) Late Cretaceous IT and HT overprints (●) of normal polarity (Table 7*). (f) Early Permian overprints (●) predominantly of reverse polarity (Table 7*). Error circles and error ellipses, or their envelope, are shown.

the overprints with IT or soft HT blocking temperature ranges (P0, P1: 29 of 54 overprints listed). The 'smaller angle' overprints are the more obvious candidates for composite unblocking, bearing in mind the directional closeness of P1 and P2 components (Figure 14). An artefact origin is less likely for the IT and soft HT overprints with larger angular distances, representing the majority, and is unlikely for the hard HT overprints (P2: 25 of 54).

Conglomerate tests

Component directions of volcanic pebbles and matrix have been compared for conglomerate sites 48 and 58. Both sites show distributed HT directions for the volcanic pebbles [site 48: k = 1.1, α_{95} = 141.2°, R = 1.85, n = 8, R_o (8, 99%) = 5.26; site 58: k = 2.4, α_{95} = 40.5°, R = 6.23, n = 10, R_o (10, 99%) = 5.94] and far better to well-grouping HT directions for the matrix, identified as Permo-Triassic overprints (Table 7*). LT directions close to the present local field are dominant in the matrix and less so in the volcanic pebbles. The results demonstrate a positive conglomerate test (site 48: likely randomness at the 99% confidence level) and capability of the volcanic pebbles to retain an original magnetisation under conditions leading to pervasive resetting of the matrix.

Primary magnetisations

Thermal demagnetisations have been continued routinely until magnetic intensities reduced below the measurement limit of the ScT cryogenic magnetometer or, for a few sites, until development of viscous magnetic phases impeded remanence measurements. The demagnetisations show that separation of magnetic components generally continues until high to very high temperatures with subtle, but systematic and mainly declinational, changes, and that detailed analysis is required to distinguish primary components from incompletely cleaned pseudo-components. The Zijderveld plots show the general presence of two (P1, P2), occasionally three (P0, P1, P2), HT 'components' of successively harder thermal characteristics (Tables 2, 5*) that are removed in magnetite and hematite unblocking temperature ranges between 400°C and 700°C. The magnetically hardest P2 components (Table 2) consistently show a shift toward more westerly declinations than for the P1 components with little change in inclination (Figure 14a; Table 3). Mean site results for the P2 components show better grouping (Figure 14b; Table 3), improved positive fold tests (Table 3) and slightly lower mean inclinations (Figure 14c; Table 3), compared with the P1 'components' (Tables 5*, 6*). Therefore, the P2 component has been taken as the preferred, more precise, representation of the primary magnetisation, although a minority has been interpreted as overprints (Table 7*). The P1 and P0 'components' with lower unblocking temperature ranges have generally been interpreted as incompletely cleaned pseudo-components (Table 5*) despite their convincingly positive fold tests (Table 6*), or as overprints (Table 7*).

CARODA FORMATION

High Valley andesite tuff (site 1)

The P2 component is carried by hematite with a blocking temperature range between 570°C and 680°C. Its pole position (a on Figure 15; Table 4) locates close to preliminary poles for the upper Isismurra Formation of the Rouchel Block (Klootwijk 1996b figure 12c; 1996a table 1.3). There are no SHRIMP dates available for the High Valley andesite tuff nor for the upper Isismurra poles, but this resemblance is in agreement with available stratigraphic control. The P2 component is interpreted as the primary magnetisation for the High Valley andesite tuff and, if confirmed, bears special importance. It is the oldest pole determined here for

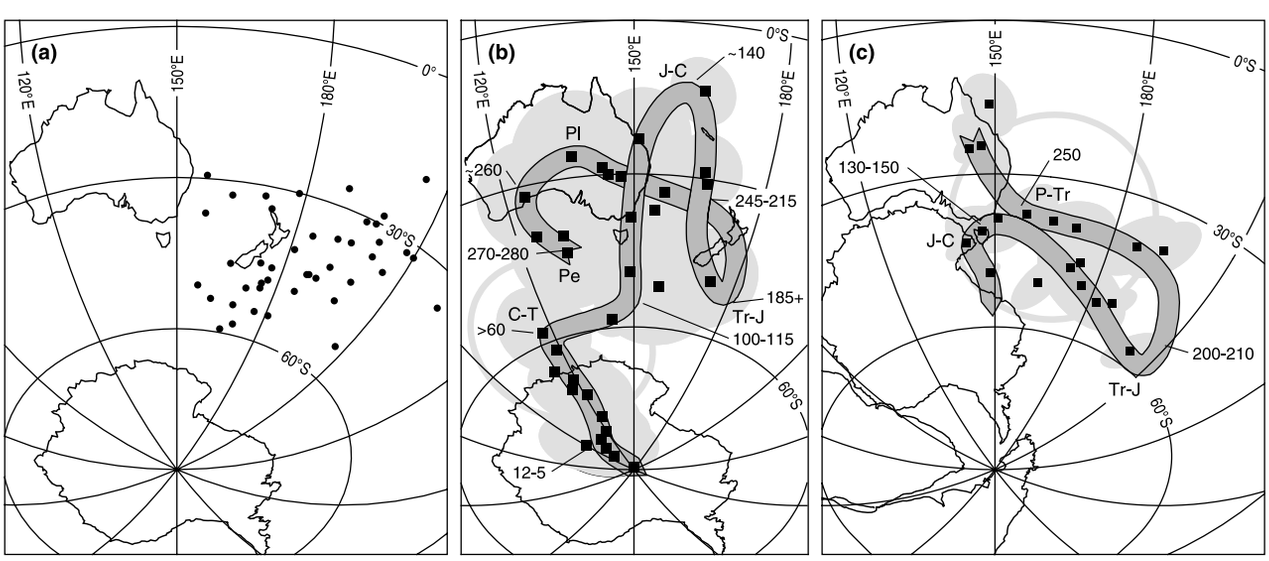


Figure 13 (a) South pole positions for pre-folding IT and HT overprints of probable Permo-Triassic origin (●: Table 7*). (b) Australian Late Palaeozoic – Cenozoic pole path, compiled from data in Klootwijk (1996a tables 1.1, 1.2) and for the Late Cretaceous and Cenozoic from Idnurm (1994) and Klootwijk (in press a figure 8), approximate ages according to Klootwijk (1996b). (c) Indian latest Palaeozoic – Mesozoic pole path transferred to Australia, with India shown in a Gondwanaland reconstruction anchored to Australia in its present position.

the Rocky Creek Block and is the only result indicating a northern hemisphere palaeolatitude (6°N). Poles for the P1 and P0 component of the High Valley andesite tuff locate close to P2 poles for sites of the presumably younger Yuendoo Rhyolite (Figure 16a–c), but are not interpreted as related. The P1 pole locates close to P2 poles for sites 7 and 9 of the Yuendoo Rhyolite (Tables 2, 5*; Figure 16a–c), but their cleaning is thought incomplete (see below). The P1 component is interpreted as a pseudo-component. The P0 pole locates close to the main pole for the Yuendoo Rhyolite, but also near to the Permo-Triassic track (Table 7*; Figure 13) and is interpreted as a Permo-Triassic overprint.

Yuendoo Rhyolite (sites 5–10)

The P2 components for three sites (5, 6, 8) from both limbs of the Slaughterhouse Syncline (Figure 4; Table 1) show a well-defined mean pole position (1 on Figure 15; Table 4) with a significantly positive fold test at the 99% confidence level (Table 3) (McFadden 1990), and are taken as the primary magnetisation. Sites 7 and 9 with MD magnetite characteristics, show south pole positions away from this main grouping and deflected toward the south pole (Figure 16c), possibly reflecting incomplete removal of a reverse Recent (Holocene) field component. The distributed HT directions of site 10 are probably of lightning origin. The primary pole (P2) for the Yuendoo Rhyolite is crucial to definition of the Visean part of the Rocky Creek pole path and ultimately to interpretation of rotational deformation of the Tamworth Belt. There is good potential to study additional east-dipping sites, but far less potential for west-dipping sites with suitable outcrop apparently limited to the studied site on the eastern limb of the Slaughterhouse Syncline (Figure 4). Primary magnetisations could not be identified for several tuffaceous horizons below the Yuendoo Rhyolite (sites 2–4, Table 7*).

West Lynne Rhyodacite (sites 11–13)

Pole positions for these three sites disperse upon bedding correction and no representative primary magnetisation direction or pole position could be identified (see Appendix 1*).

Ermelo Pyroclastics (sites 14–21)

Two sites only, 14 and less clearly 15 (Table 2), show an interpretable, rather soft, magnetite component of normal polarity that could be taken as a primary magnetisation for reason that its, ill-defined, pole locates close to, and on the younger side of, the well-defined pole for the Yuendoo Rhyolite.

CLIFDEN FORMATION AND ROCKY CREEK CONGLOMERATE

Tuff below the Glen Idle Rhyolite (site 22)

Pole positions for the HT components (P0–P2) for this site fall on a great circle with the pole for the harder (500–580°C) P2 component close to the well-defined pole position for the Yuendoo Rhyolite. The IRM acquisition curve indicates the presence of MD magnetite. In the absence of a fold test it cannot be established whether this harder P2 component represents the primary magnetisation or a composite.

Peri Rhyolite (sites 26, 27)/Boomi Rhyolite (site 30) and associated tuffs (sites 24, 31–33)

Magnetic component directions, rock-magnetic properties (IRM acquisition, Lowrie test) and magnetic fabric results for the Boomi Rhyolite and the Peri Rhyolite are comparable, underscoring their lithological similarity. Therefore, it is suggested, as a proposition testable through SHRIMP dating, that the Boomi Creek section west of Travellyn was misidentified as Caroda Formation (Hocking 1973; Stroud 1990b; Opdyke *et al.* 2000) instead of Clifden Formation, and that the Boomi Rhyolite is a more proximal equivalent of

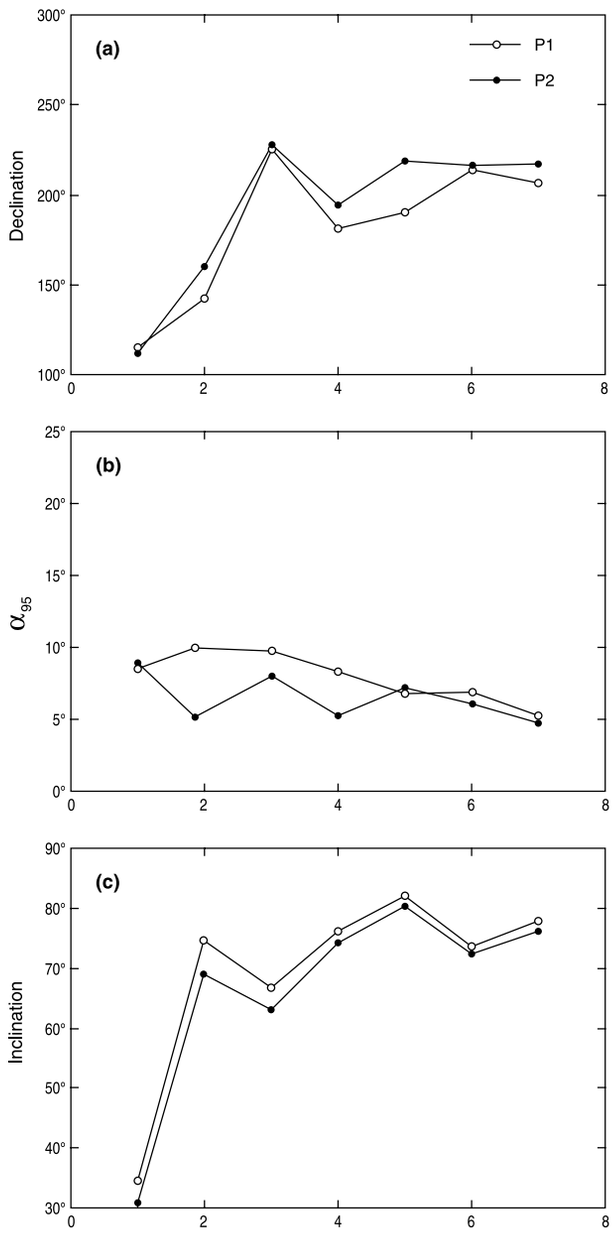


Figure 14 Comparison of directional and statistical characteristics for the HT components in the upper magnetite (P1: pseudo-components) and upper magnetite–hematite (P2: primary) unblocking temperature ranges, for the seven mean directions detailed in Table 3 (P2) and in Table 6* (P1). (a) Mean declination. (b) Half-angle of the cone of 95% confidence (α_{95}). (c) Mean inclination.

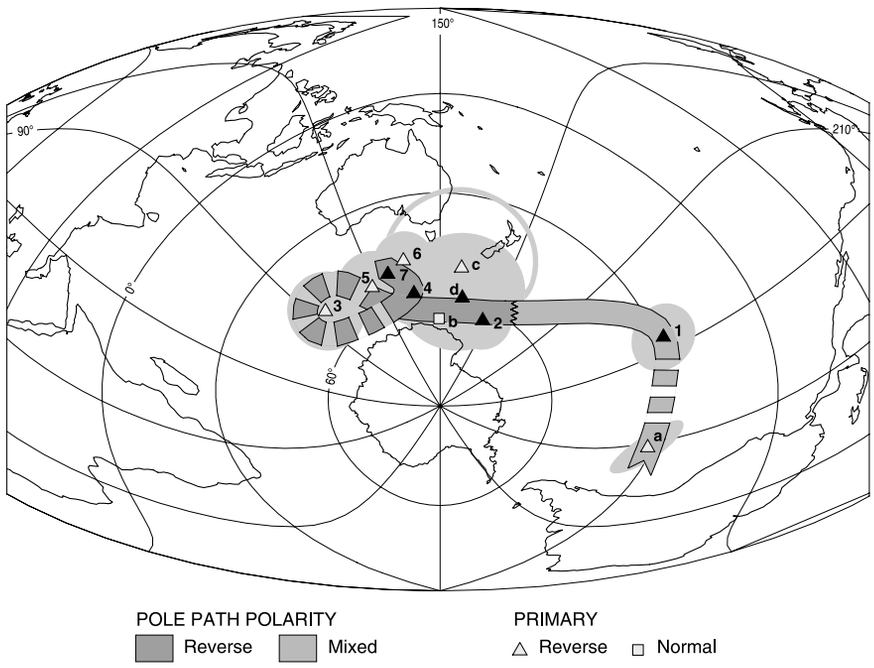


Figure 15 Carboniferous pole path for the Rocky Creek Block based on the mean pole positions detailed in Table 4. ▲, indicate the main defining poles.

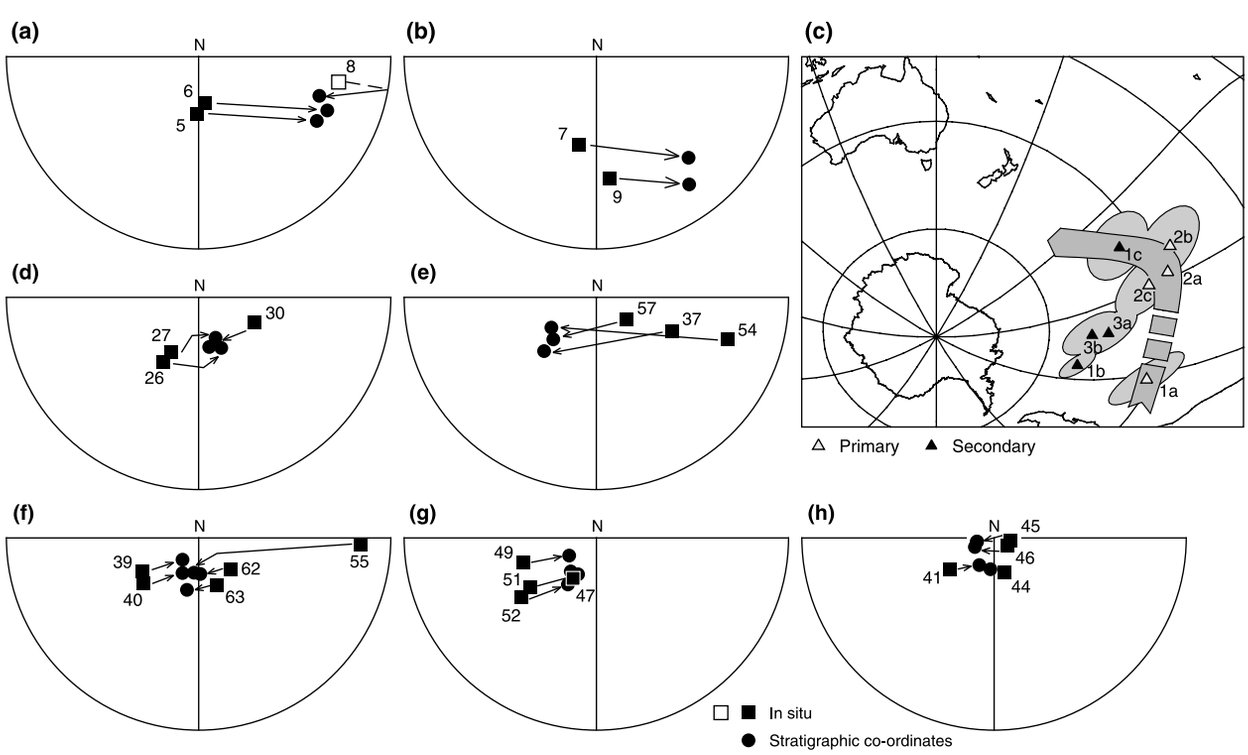


Figure 16 (a) Fold test for P2 components of Yuendoo Rhyolite sites 5, 6 and 8, identified as primary magnetisations. (b) Bedding correction for P2 components of Yuendoo Rhyolite sites 7 and 9 with suspected incomplete separation of a primary magnetisation component and a (reverse polarity) 'Recent Field'-like overprint. (c) Comparison of P2 primary components and P1 pseudo-components for the High Valley andesite tuff (site 1) and the Yuendoo Rhyolite (sites 5–9). 1a, (site) 1-P2; 1b, 1-P1; 1c, 1-P0; 2a, (site) 6-P2; 2b, 8-P2; 2c, 5-P2; 3a, (site) 7-P2; 3b, 9-P2. Compare with Figure 15. Note that pole 1b (site 1-P1) locates close to poles 3a (site 7-P2) and 3b (site 9-P2). Pole 1c (site 1-P0) locates close to, but on the younger side of, poles 2a–c for the Yuendoo Rhyolite, and close to Permo-Triassic overprint poles (Figure 13a). (d) Fold test for P2 components for Peri Rhyolite sites 26, 27 and 30. (e) Fold test for P2 components for upper Clifden Formation – lower Rocky Creek Conglomerate sites 37, 54, 57. (f) Fold test for P2 components for Rocky Creek Conglomerate sites 39, 40, 55, 62, 63. (g) Fold test for P2 components for Lark Hill Formation (Tops Road) sites 47, 49, 51, 52. (h) Fold test for P2 components for Lark Hill Formation (Rocky Creek) sites 41, 44–46.

the Peri Rhyolite. Combination of the well-defined P2 directions for the three sites (26, 27, 30) provides a well-defined pole (2 on Figure 15, Figure 16d; Table 4) with a positive fold test (Table 3) interpreted to represent the primary magnetisation. Inclusion of the less well-defined P2 direction for a stratigraphically slightly lower tuff (24) does not improve the pole's statistics and is not preferred as the more inclusive representative pole. The mean pole for the P2 components (480–580°C: Table 4) from three thin tuffaceous horizons (31–33) above the Boomi Rhyolite locates to the west of the Peri Rhyolite – Boomi Rhyolite pole, supporting east-to-west younging of the pole path (d on Figure 15).

Upper Clifden Formation – lower Rocky Creek Conglomerate (sites 37, 53, 54, 57)

Poles for the P2 components for the four sites locate to the south-southwest of Australia, with the poorly defined pole for site 53 located away from the more clustered poles for the other three sites (3 on Figure 15; Table 2). Notably these three sites come from three different tectonic units—Kathrose, Rocky Creek and Darthula Thrust Sheets—and show a well-defined mean pole (Table 4) with a significantly (99%) positive fold test (Figure 16e; Table 3).

Rocky Creek Conglomerate (sites 39, 40, 55, 62, 63)

These sites from three different thrust sheets, Rocky Creek (39,40), Kathrose (55) and Darthula (62,63), show well-grouped poles with a significantly (95%) positive fold test (Figure 16f; Table 3) and a mean pole (4 on Figure 15; Table 4) to the south of the Great Australian Bight. Good grouping of poles for sites from the three thrust sheets, demonstrated by the results for the Rocky Creek Conglomerate and the upper Clifden Formation – lower Rocky Creek Conglomerate, indicates absence of significant rotational deformation between the thrust sheets.

Lark Hill Formation (sites 41, 44–47, 49, 51, 52)

P2 components for these sites show well-grouped poles (Table 4) with significantly (99%) positive fold tests (Figure 16g, h; Table 3), and an overall mean-site pole to the south of the Great Australian Bight (7 on Figure 15; Table 4), but closer inshore than the pole for the Rocky Creek Conglomerate. There is, however, a systematic difference between poles for the Rocky Creek section and for the Tops Road with the mean-site pole for the latter section closer to the pole for the Rocky Creek Conglomerate and for the former closer to the Great Australian Bight section (Tables 3, 4; 5 and 6 on Figure 15). In the absence of detailed isotopic or palaeontological dating control it is unclear whether these two poles represent different stratigraphic levels and possibly a pole path trend. The overall mean-site pole is used in pole path construction.

Polarity

The main objective of this study is to define the Carboniferous pole path for the Tamworth Belt. Magnetostratigraphic constraints on the base of the Permo-Carboniferous Reverse Superchron are a fortuitous by-product. Primary palaeomagnetic results show exclu-

Table 4 Pole path data.

Pole no.[a]	Formation/site Formation/Member	Site No.	*n*	Direction Decl (°)	Incl (°)	KD	ED95 (°)	Lat (°S)	Long (°E)	South Pole position Palaeolat (°S)	Palaeolat[b] (°S)	DP (°)	DM (°)	KP	EP95 (°)
a	High Valley andesite tuff	1	1	132.5	–13.8	25.4	8.8	31.3	271.4	7.0N	5.8N	4.6	9.0	–	–
1	Yuendoo Rhyolite	5,6,8	3	112.6	31.0	189.0	9.0	27.7	235.8	18.1	18.3	–	–	184.0	9.1
b	Kooringal Dacite (CICA01-05) (Clifden Fm?)	29	1	0.6	–70.6	451.8	2.2	65.3	149.5	54.8	55.2	3.2	3.7	–	–
c	Site AIPA, ~5 m below Peri Rhyolite	24	1	164.3	78.6	20.1	11.9	50.8	159.5	68.1	68.9	21.1	22.2	–	–
2	Peri Rhyolite	26,27,30	3	160.4	69.0	568.8	5.2	63.4	177.4	52.5	53.3	–	–	228.5	8.2
d	Tuffac. beds above Boomi Ryolite (Clifden Fm?)	31–33	3	167.8	73.9	147.7	10.2	59.1	162.2	60.0	60.5	16.5	18.3	–	–
3	Upper Clifden Fm.- lower Rocky Creek Congl.[c]	37,54,57	3	228.2	63.1	231.3	8.1	49.6	95.3	44.6	44.1	–	–	127.5	11.0
4	Rocky Creek Conglomerate	39,40,55,62,63	5	195.0	74.3	210.5	5.3	57.6	136.5	60.7	60.9	–	–	72.7	9.0
5	Lark Hill Fm. (Tops Road)	47,49,51,52	4	217.0	72.3	224.7	6.1	52.1	118.4	57.5	57.5	–	–	81.4	10.2
6	Lark Hill Fm. (Rocky Creek)	41,44–46	4	219.0	80.5	159.4	7.3	48.0	134.8	71.5	68.3	–	–	123.0	8.3
7	Lark Hill Fm. Combined	41,44–47,49,51,52	8	217.7	76.4	132.7	4.8	50.4	127.0	64.2	62.9	–	–	70.2	6.7

[a]See Figure 15 for pole path. Numbers indicate results with positive fold test (see Table 3), letters indicate single site results or multi-site results with uniform bedding attitude (no fold test).
[b]Armidale (151.7°E, 30.5°S) reference location.
[c]Questionable primary origin of magnetisation.
K (KD, KP), best estimate of the precision parameter κ (Fisher 1953); DP, DM, half-angles of the oval of 95% confidence around the mean pole; ED95, (EP95), half-angle of the circle of 95% confidence around the mean direction (mean pole).
n = number of sites

sively reverse polarity (Table 2) with two exceptions: the Ermelo Pyroclastics, with a tentatively identified normal polarity (sites 14, 15); and the Kooringal Dacite (29). These observations agree with the extensive magnetostratigraphic study of the Rocky Creek and Werrie Block successions by Opdyke *et al.* (2000) (summarised in Figure 5), who successfully located the older boundary of the Permo-Carboniferous Reverse Superchron within the Pinnacles section of the southern Rocky Creek Syncline (Opdyke *et al.* 2000 figures 4, 5, 11) to the south of the region studied here, between the normal polarity Wanganui Andesite (319.2 ± 2.8 Ma) and the reverse polarity Eastons Arm Rhyolite (317.8 ± 2.8 Ma). There is a qualifier to this location: a normal polarity ignimbrite apparently overlies the reverse polarity Wanganui Andesite, but structural complexity clouds its magnetostratigraphic significance. It could: (i) represent a normal polarity zone in the basal part of the Permo-Carboniferous Reverse Superchron; (ii) require redefinition of the base of the Permo-Carboniferous Reverse Superchron above this ignimbrite; or (iii) represent a stratigraphic misidentification. Opdyke *et al.* (2000) correlated the Eastons Arm Rhyolite with the Peri Rhyolite (318.0 ± 3.4 Ma) and evidence is presented here that the Peri Rhyolite can be correlated with the Boomi Rhyolite. The present results contribute to further location of the Permo-Carboniferous Reverse Superchron base constraining it between the Ermelo Pyroclastics [normal polarity (N): 321.3 ± 3.2 Ma] and the Peri Rhyolite [reversed polarity (R): 318.0 ± 3.4 Ma], and also between the Kooringal Dacite (N) and the Boomi Rhyolite (R) in the Boomi Creek section, which has potential for further refining the location.

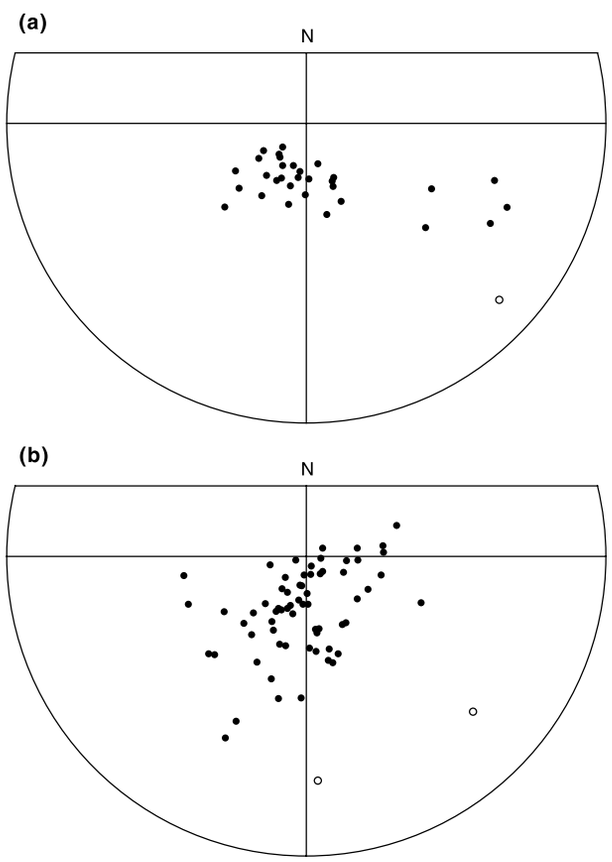


Figure 17 Comparison of site-mean directions in equal-area projection for (a) P2 components for Rocky Creek Block sites listed in Table 2 and (b) B-components for Rocky Creek Block sites after Opdyke *et al.* (2000 table 1: site-mean results shown for *n* (samples) ≥3). ●, lower hemisphere; ○, upper hemisphere.

DISCUSSION

Comparison with regional palaeomagnetic results

Opdyke *et al.* (2000) interpreted the directional results of their extensive, excellently executed, magnetostratigraphic study on the Carboniferous succession of the Rocky Creek and Werrie Blocks in terms of pole positions. Two of their conclusions differ from the interpretations presented here and warrant commenting. These are: (i) the scatter in declinations for Caroda Formation sites of the lower succession implies the possibility of modest relative rotations between sites; and (ii) mean directions for formations of the upper succession–Clifden Formation, Rocky Creek Conglomerate and Lark Hill Formation of the Rocky Creek Block and the Currabubula Formation of the Werrie Block–are remarkably uniform.

The thermal demagnetisation procedures followed by Opdyke *et al.* (2000) are more than adequate for magnetostratigraphic purposes, although not aimed specifically at analysing the hardest few percent of magnetisation. They identified two magnetisation components: a Recent Field component 'A' with low unblocking temperatures, and a primary component 'B'. Unblocking temperatures for the latter commonly range from 400°C to 580°C, but extend sometimes to above 600°C, with the same directions for magnetite and hematite contributions. The present results, in contrast, show subtle, but consistent, differences between the harder P2 components and the softer P1 pseudo-components, with marked improvement in accuracy of mean-site directions (Figure 14b; Table 4), although not necessarily so for the confidence level of the significantly positive fold tests (Tables 3, 6*) for P2 versus P1.

The spread in site-mean directions is considerable for both Opdyke *et al.*'s (2000) study and the present study (Figure 17), with a somewhat wider spread in Opdyke *et al.*'s results. Their study did not specifically address whether this spread may reflect a directional evolution through the succession. The positive fold tests (Table 3) obtained in the present study for individual volcanic horizons, stratigraphic sections and formations demonstrate that the spread reflects an east-to-west younging pole path structure (Figure 15). This structure is recognisable also in Opdyke *et al.*'s (2000 table 3) mean pole positions, albeit to a lesser extent, and presents a robust alternative to their view [conclusions (i) and (ii) above] on possible local rotations expressed in the lower succession and on directional uniformity through the upper succession.

Pole path

The Carboniferous pole path for the Rocky Creek Block emerging from this study is very long (Figure 15) and indicates large-scale rotational and latitudinal movements for the block. The newly defined poles further detail the main

features of the preliminary pole path for the Tamworth Belt, based on pilot results (Klootwijk 1996b figure 13). However, the new poles for the upper part of the studied succession, that is, for the upper Clifden Formation to the Lark Hill Formation, would require a substantial change to the shape of the Late Carboniferous westward loop on the preliminary pole path, if the pole for the upper Clifden Formation – lower Rocky Creek Conglomerate (3 on Figure 15) is indeed taken as primary. This would imply that the younger limb of the westward loop does not follow a west-to-east track through the Great Australian Bight, but would double onto itself at more southern latitudes in Rocky Creek Conglomerate time (4 on Figure 15) before moving northward toward the Great Australian Bight in Lark Hill Formation time (7 on Figure 15). Pole path loops carry an inherent danger of unrecognised overprints. The three upper Clifden Formation – lower Rocky Creek Conglomerate sites (37, 54, 57) are all sampled close to major fault zones and potential overprinting of pre-folding (pre-Middle Triassic) origin, facilitated by fluid flow along the fault zones, cannot be ignored. Pending further definition of Late Carboniferous poles for the correlatable Currabubula Formation in the southward adjacent Werrie Block, a primary origin for the upper Clifden Formation – lower Rocky Creek Conglomerate pole, and with it the reality of a Late Carboniferous loop, may be questioned.

Pole positions for the underlying succession, the lower Clifden Formation and the Caroda Formation, further detail the Early Carboniferous eastward loop (Figure 15) outlined on the preliminary pole path (Klootwijk 1996b figure 13). The Peri Rhyolite and Yuendoo Rhyolite poles (2 and 1 on Figure 15), in particular, detail the shape of the younger limb of the eastward loop. The pole position for the High Valley andesite tuff from the lower part of the Caroda Formation (a on Figure 15) may provide an important backward extension to the pole path for the Rocky Creek Block. This pole position falls on the older limb of the eastward loop and is virtually indistinguishable from preliminary pole positions for the upper part of the Isismurra Formation of the Rouchel and Gresford Blocks (Klootwijk 1996a; Klootwijk 1996b figure 13). Implications of this observation for rotational deformation of the Tamworth Belt will be explored in another paper.

CONCLUSIONS

Palaeomagnetic, rock magnetic and magnetic fabric studies of volcanic and volcaniclastic horizons in the Carboniferous succession (Caroda Formation – Lark Hill Formation) of the Rocky Creek Block have shown:

(1) No noticeable rotation between the three major thrust sheets—the Rocky Creek, Kathrose and Darthula—of the Rocky Creek Block.

(2) Consistent transport directions in the older succession, Caroda Formation to Clifden Formation, and more variable transport patterns in the younger succession, Rocky Creek Conglomerate and Lark Hill Formation.

(3) Four periods of magnetic overprinting: widespread overprinting of mid-Tertiary (perhaps Recent viscous) and Permo-Triassic origin, and more restricted Late Cretaceous and Early Permian overprinting.

(4) Well-defined primary magnetisation components in all studied formations—Caroda Formation, Clifden Formation, Rocky Creek Conglomerate and Lark Hill Formation—apart from the Ermelo Pyroclastics. The primary components reside in magnetite and/or hematite carriers.

(5) Subtle, but systematic, directional differences between softer magnetic components residing mainly in the magnetite fraction and harder components residing mainly in the hematite fraction. The latter, with systematically smaller α_{95} values, have been interpreted as primary magnetisation components, the former as incompletely cleaned pseudo-components.

(6) A long Visean to Westphalian, east-to-west younging, pole path trajectory supported by at least four positive fold tests, three with confidence levels above 95% or 99%.

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SUPPLEMENTARY PAPERS

Appendix 1 Palaeomagnetic results—formation and site comments.
Table 5 Pseudo-components (HT) in stratigraphic coordinates.
Table 6 Fold test after McFadden (1990) for HT pseudo-components.
Table 7 Intermediate- temperature (IT) and high-temperature (HT) overprints in geographical (G) or stratigraphic (S) coordinates.